

FARM MANAGEMENT SYSTEM

User Story:

In the Farm Management System (FMS), as a farmer, farm manager, or system administrator, I want to use a centralized Salesforce-based platform to record and manage farm, crop, farmer, and buyer information so that all agricultural data is stored securely and systematically. I want to enter and update farm details, crop production records, and buyer transactions easily through a user-friendly interface with proper validation rules to prevent incorrect entries. I also want the system to generate real-time reports and interactive dashboards so that I can analyze farm productivity, monitor sales, and make informed decisions. Additionally, I want role-based access and secure data storage to ensure that sensitive information is protected while enabling smooth collaboration among users. This system will help reduce manual paperwork, minimize errors, improve efficiency, and support data-driven farm management

1. INTRODUCTION

1.1 Project Overview :

The Farm Management System (FMS) is a Salesforce-based application designed to digitize and streamline agricultural operations. It manages data related to farms, crops, farmers, and buyers while providing real-time insights through reports and dashboards. The system helps farmers and managers monitor agricultural activities efficiently and make informed decisions.

1.2 Purpose :

The main purpose of this project is to:

- Digitize agricultural data management
- Improve data accuracy and accessibility
- Automate record handling
- Enable real-time monitoring

- Support data-driven farming decisions
- Enhance productivity and transparency

2. IDEATION PHASE

2.1 Problem Statement (Customer Problem Statement Template)

PS	I am (Customer)	I'm trying to	But	Because	Which makes me feel
PS-1	Farmer	Maintain farm and crop records	Data is often incomplete or incorrect	Records are kept manually in registers	Frustrated and confused
PS-2	Farm Manager	Monitor farm productivity and sales	Reports are delayed and inaccurate	Data is scattered across files and sheets	Stressed and inefficient
PS-3	Buyer	Access crop availability and pricing	Information is not updated regularly	There is no centralized digital system	Uncertain and dissatisfied
PS-4	System Admin	Manage user access and system security	Unauthorize d access may occur	Access control is handled manually	Concerned and insecure

Problem Statements – Farm Management System

PS	I am (Customer)	I'm trying to	But	Because	Which makes me feel
PS-1		Maintain farm and crop records	Data is often incomplete or incorrect	Records are kept manually in registers	
PS-2	 Farm Manager	Monitor farm productivity and sales	Reports are delayed and inaccurate	Data is scattered across files and sheets	 Stressed and inefficient
PS-3		Access crop availability and pricing	Information is not updated regularly	There is no centralized digital system	 Uncertain and dissatisfied
PS-4	 System Admin	Manage user access and system security	Unauthorized access may occur	Access control is handled manually	 Concerned and insecure

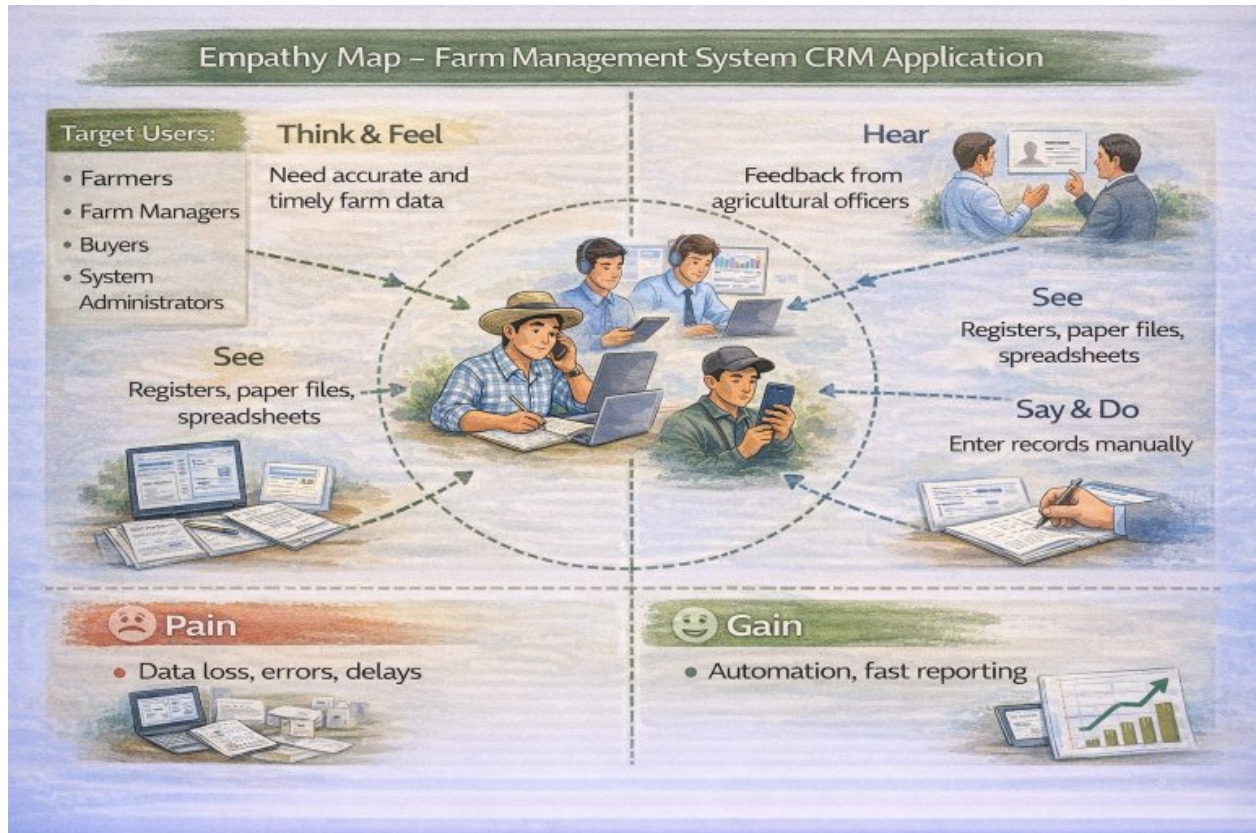
2.2 Empathy Map Canvas

Target Users:

- Farmers
- Farm Managers
- Buyers
- System Administrators

Empathy Analysis

- **Think & Feel:** Need accurate and timely farm data
- **Hear:** Feedback from agricultural officers
- **See:** Registers, paper files, spreadsheets
- **Say & Do:** Enter records manually
- **Pain:** Data loss, errors, delays
- **Gain:** Automation, fast reporting



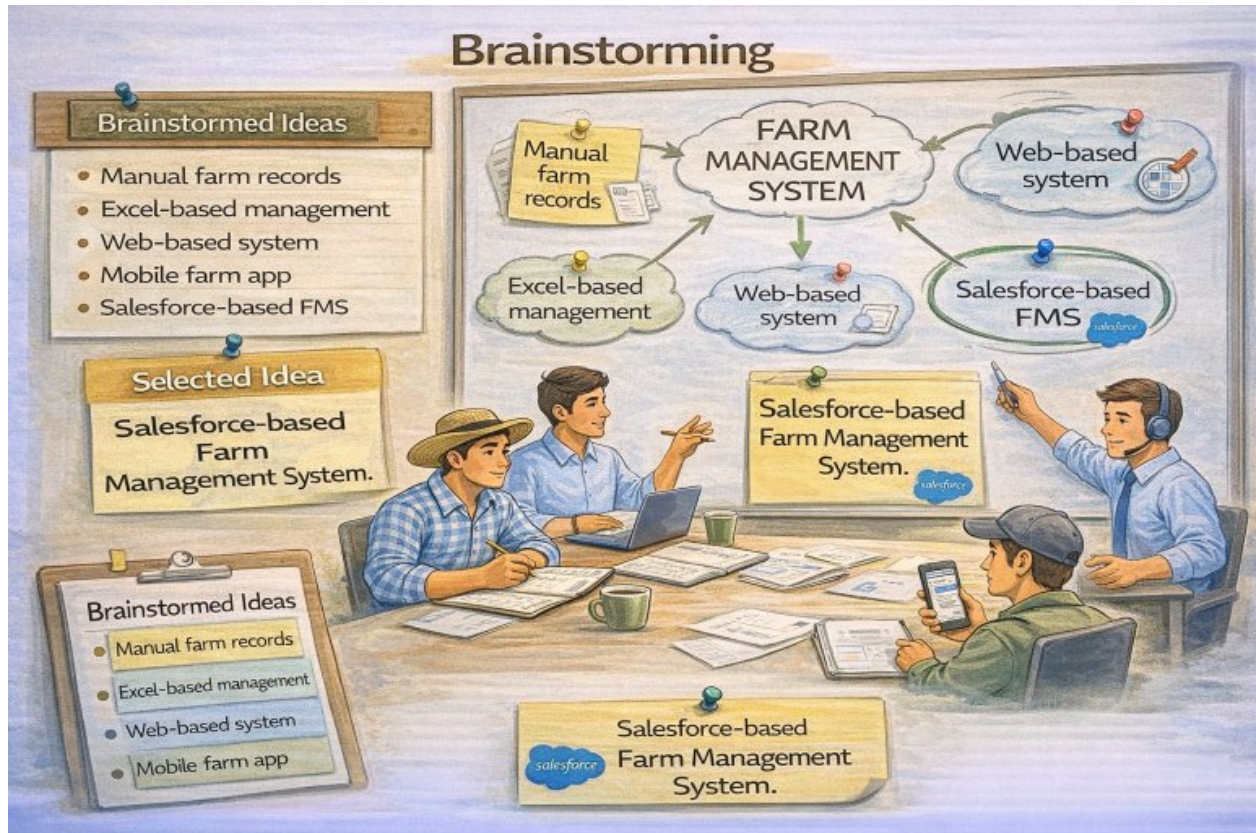
2.3 Brainstorming

Brainstormed Ideas

- Manual farm records
- Excel-based management
- Web-based system
- Mobile farm app
- Salesforce-based FMS

Selected Idea

Salesforce-based Farm Management System.

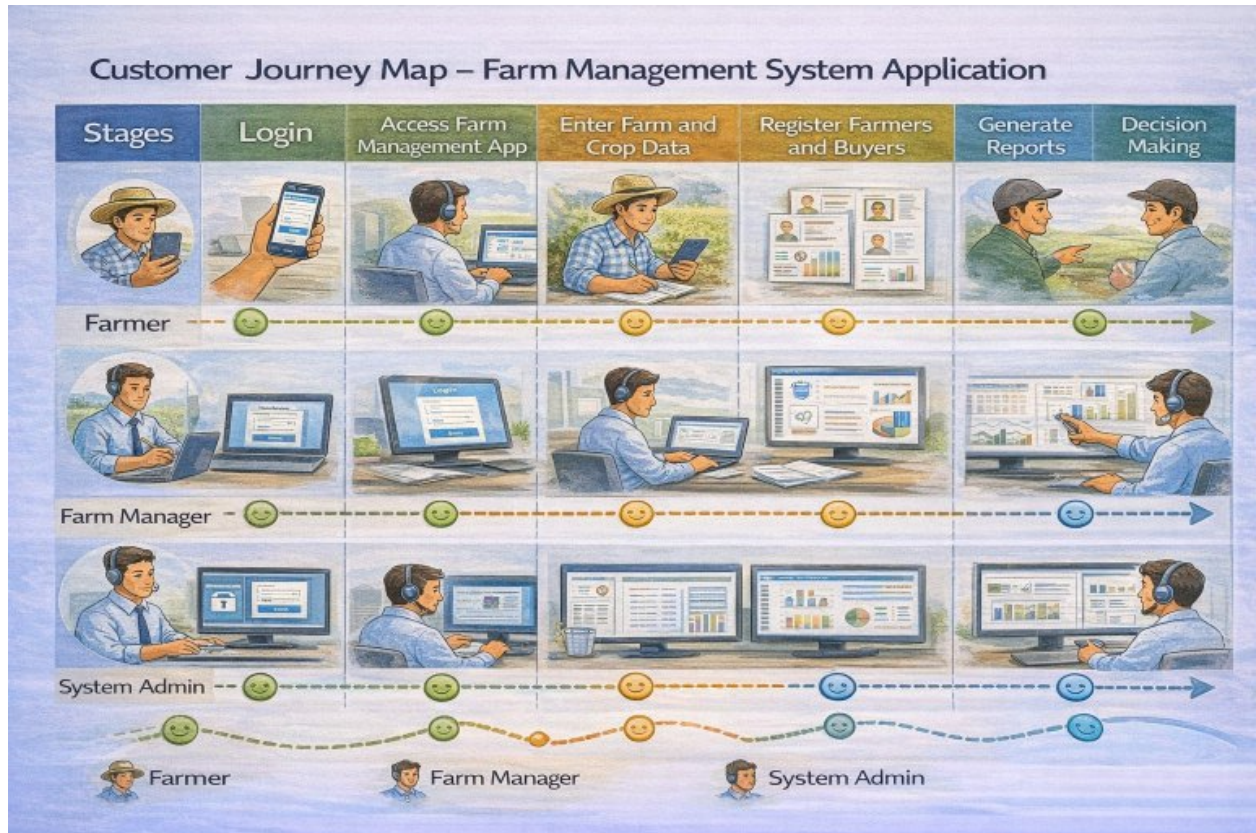


3.REQUIREMENT ANALYSIS PHASE

3.1 Customer Journey Map

Stages:

1. Login
2. Access Farm Management App
3. Enter Farm and Crop Data
4. Register Farmers and Buyers
5. Generate Reports
6. Analyze Results
7. Decision Making



3.2 Solution Requirements

Functional Requirements

FR No	Requirement
FR-1	User Authentication
FR-2	Farm, Crop, Farmer, Buyer Management
FR-3	Data Entry and Validation
FR-4	Relationship Mapping
FR-5	Report Generation
FR-6	Dashboard Visualization

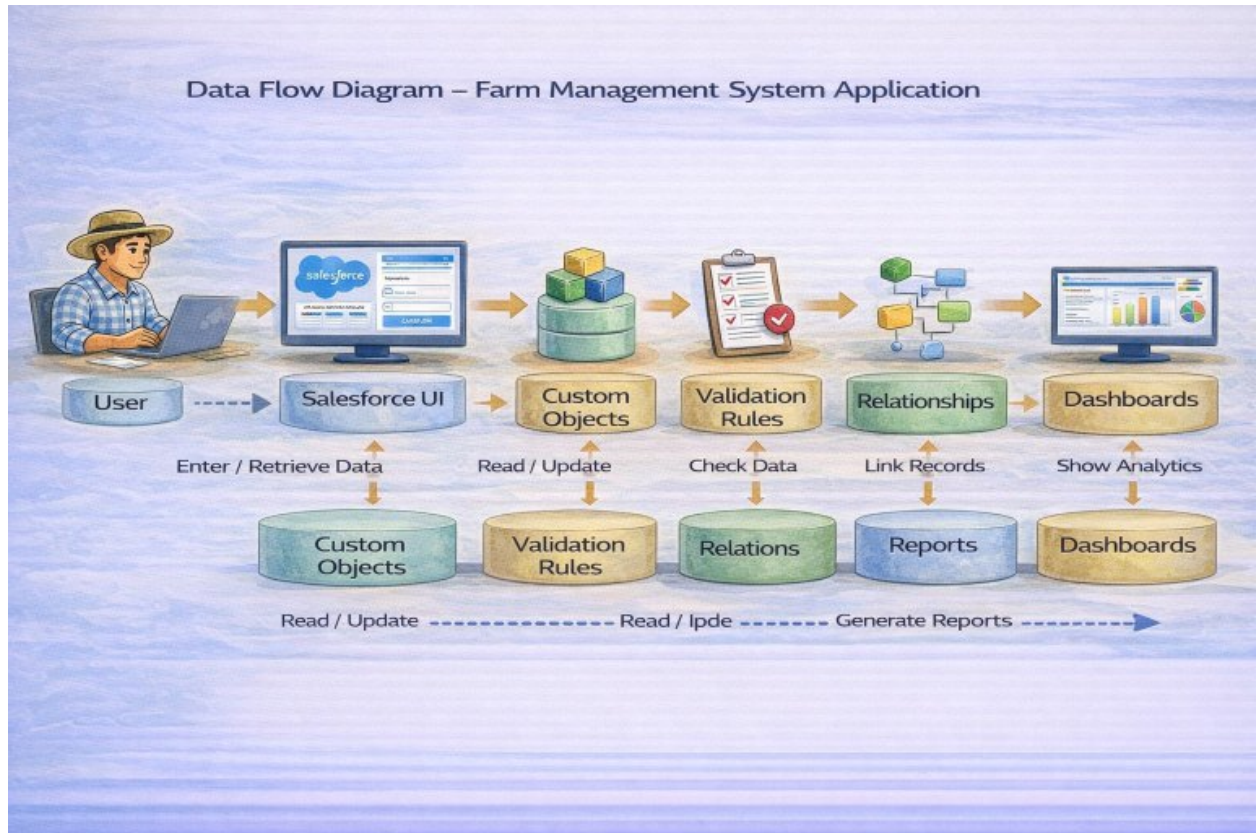
Non-Functional Requirements

NFR	Description	Example
Usability	Easy interface	Simple Lightning UI
Security	Data protection	Profiles & roles
Reliability	No data loss	Cloud backup
Performance	Fast access	Reports in <3 sec
Availability	24/7 access	Cloud hosting
Scalability	Handles growth	More farms/users

3.3 Data Flow Diagram

Data Flow:

User → Salesforce UI → Custom Objects (Farm, Crop, Farmer, Buyer) → Validation Rules → Relationships → Reports → Dashboards



3.4 Technology Stack

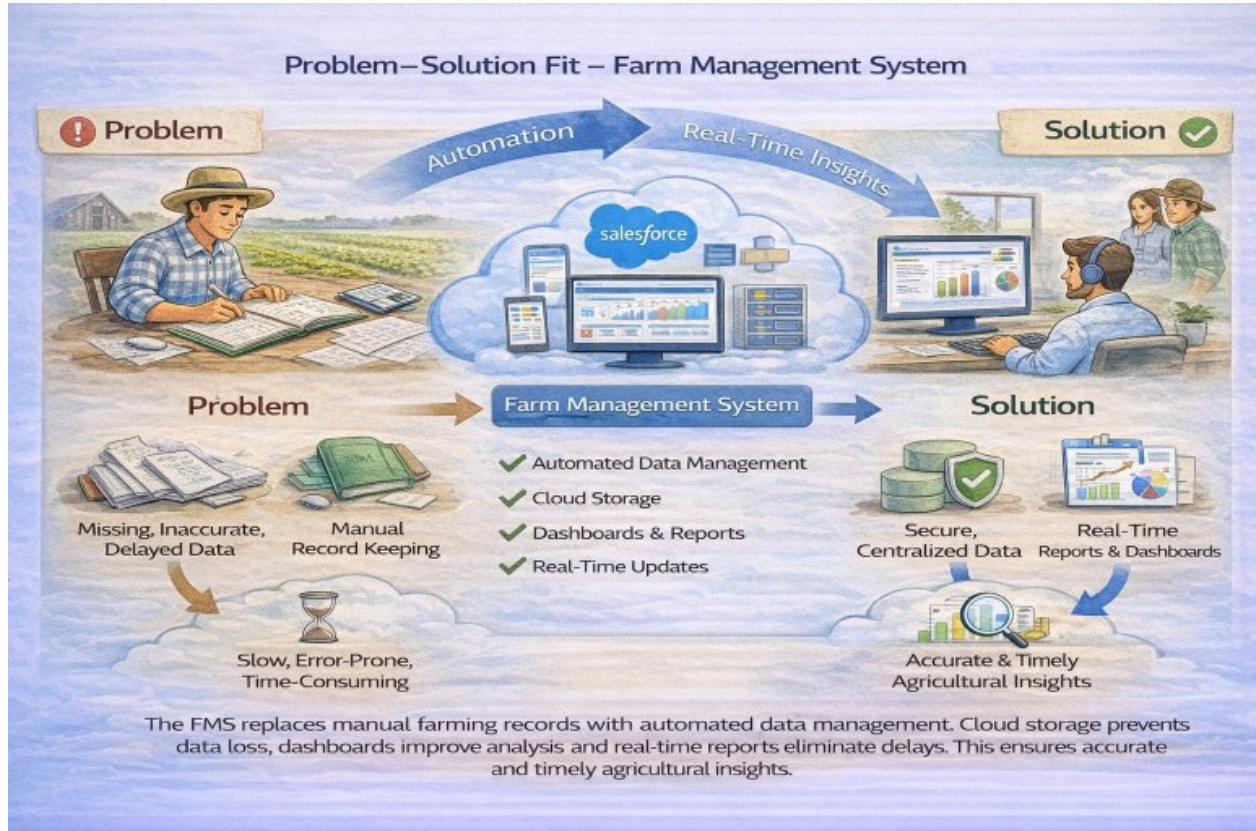
Layer	Technology
Frontend	Salesforce Lightning UI
Backend	Salesforce Platform
Database	Salesforce Cloud
Automation	Validation Rules
Analytics	Reports & Dashboards

4. PROJECT DESIGN PHASE

4.1 Problem–Solution Fit

The FMS replaces manual farming records with automated data management. Cloud storage prevents data loss, dashboards improve analysis, and real-time reports eliminate delays. This ensures accurate and

timely agricultural insights.



4.2 Proposed Solution

Functional Requirements

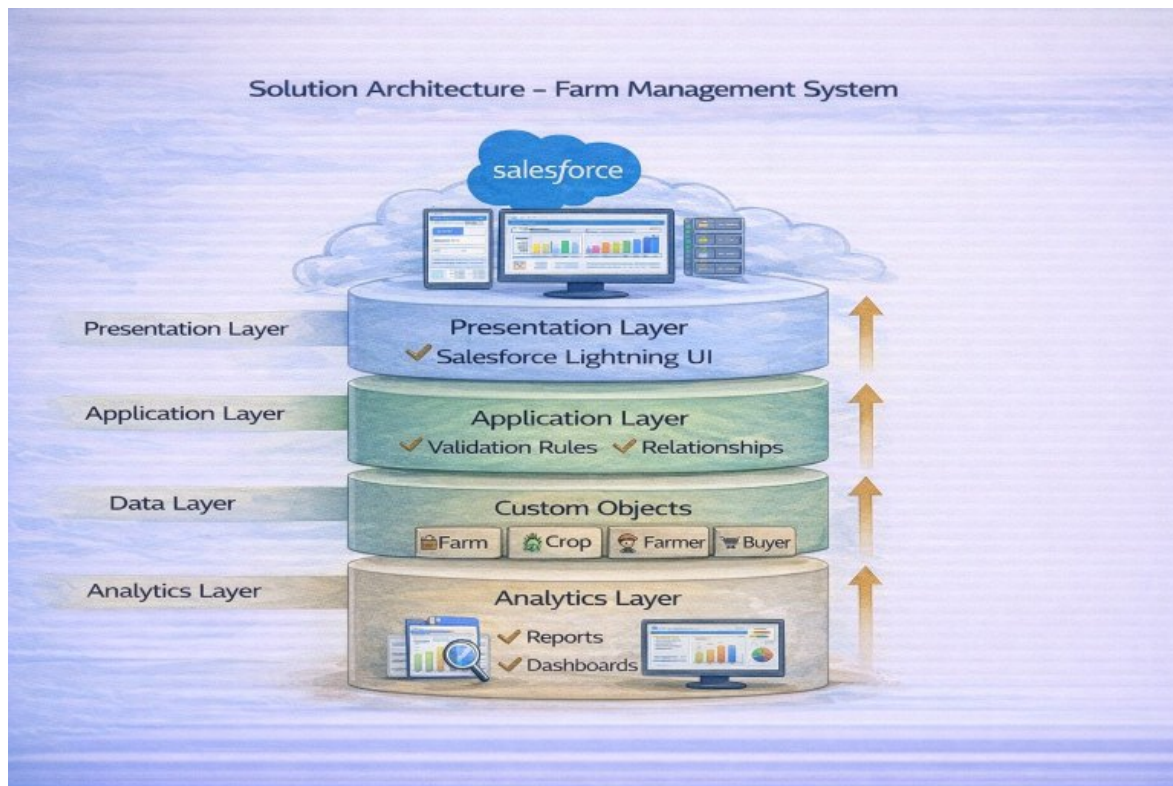
Parameter	Description
Problem	Manual maintenance of farm, crop, farmer, and buyer records leading to errors, data loss, and delays.
Solution	Salesforce-based Farm Management System with centralized data storage and automation.
Innovation	Automated data entry validation, cloud storage, and real-time reporting dashboards.
Impact	Improved data accuracy, faster reporting, and better decision-making for farmers and managers.
Business Model	Digital farm management and agricultural data services platform.

Scalability	Supports multiple farms, users, and regions with cloud-based infrastructure.
-------------	--

4.3 Solution Architecture

Architecture Layers

1. Presentation Layer – Salesforce Lightning UI
2. Application Layer – Validation Rules, Relationships
3. Data Layer – Custom Objects
4. Analytics Layer – Reports and Dashboards



5. PROJECT PLANNING & SCHEDULING

5.1 Project Planning

- Created Salesforce Developer Org
- Developed custom objects
- Configured tabs and Lightning App

- Created fields and relationships
- Designed profiles and roles
- Created users and sharing rules
- Developed reports and dashboards
- Conducted testing and reviews

Product Backlog, Sprint Schedule, and Estimation (4 Marks)

Use the below template to create product backlog and sprint schedule

Sprint	Functional Requirement (Epic)	User Story Number	User Story / Task	Story Points	Priority	Team Members
Sprint-1	Developer Setup	USN-1	As a system administrator, I want to create and configure a Salesforce developer account so that I can set up and deploy the Farm Management System.	3	High	Member1
Sprint-2a	Data Modeling	USN-2	As an admin, I want to create Farm, Crop, Farmer, and Buyer objects with proper relationships so that agricultural data is stored and managed efficiently.	5	High	Member2
Sprint-2	Data Modeling	USN-3	As a staff member, I want tabs and a Lightning App for all farm modules so that I can easily navigate and perform daily operations.	5	High	Member3
Sprint-3	Automation	USN-4	As a system, I want to validate farm and crop details such as dates, quantities, and mandatory fields so that incorrect or incomplete records are prevented.	3	High	Member4
Sprint-3	Automation	USN-5	As a manager, I want automated flows for crop updates and buyer notifications so that farm operations are processed faster without manual intervention.	3	High	Member4
Sprint-4	Security	USN-6	As an administrator, I want role-based access control for farmers, managers, and buyers so that sensitive farm and transaction data is secure.	5	High	Member2

Sprint-5	Reports	USN-7	As a farm manager, I want reports showing crop yield, sales, and buyer transactions so that I can monitor farm performance and make informed decisions.	4	High	Member3
Sprint-6	Dashboards	USN-8	As a farm manager, I want dashboards showing crop productivity, sales trends, and operational statistics so that I can monitor performance visually.	4	Medium	Member4

Project Tracker, Velocity & Burndown Chart: (4 Marks)

6. Project Development Phase:

Project Description:

The Farm Management System (FMS) in Salesforce is a comprehensive solution designed to digitize and streamline agricultural operations. This application empowers farmers, farm managers, and stakeholders by managing data related to crops, Farm, Farmer, and buyers, while providing actionable insights through reports, and dashboards.

Sprint	Total Story Points	Duration	Sprint Start Date	Sprint End Date (Planned)	Story Points Completed (as on Planned End Date)	Sprint Release Date (Actual)
Sprint-1	20	6 Days	24 Oct 2022	29 Oct 2022	20	29 Oct 2022
Sprint-2	20	6 Days	31 Oct 2022	05 Nov 2022		
Sprint-3	20	6 Days	07 Nov 2022	12 Nov 2022		
Sprint-4	20	6 Days	14 Nov 2022	19 Nov 2022		

Milestone 1-Salesforce:

What Is Salesforce?

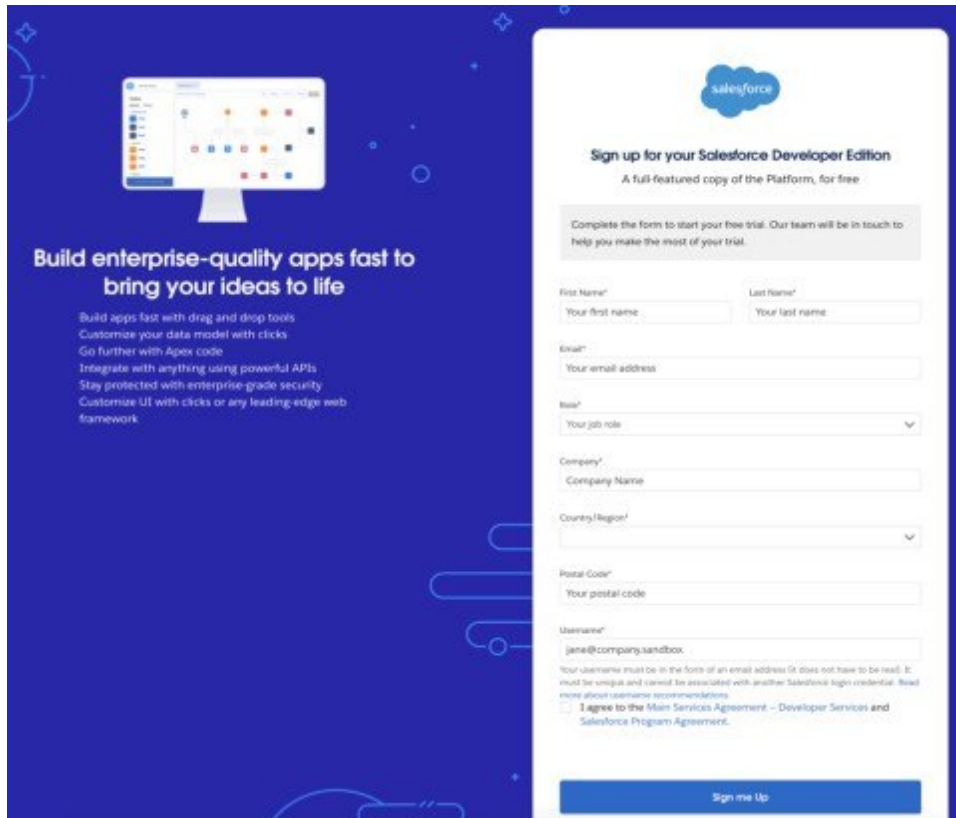
Salesforce is your customer success platform, designed to help you sell, service, market, analyze, and connect with your customers.

Salesforce has everything you need to run your business from anywhere. Using standard products and features, you can manage relationships with prospects and customers, collaborate and engage with employees and partners, and store your data securely in the cloud.

Activity1:
Creating a Salesforce Developer Org:

A Developer org has all the features and licenses you need to get started with Salesforce.

1. Search [Developer.salesforce.com/signup](https://developer.salesforce.com/signup)
2. Enter the following details like First name, last name, Email, Role, Company, Country/Region, Postal code, and Username must be unique.
3. Click sign me up, after a few min you will reserve a mail salesforce org and by using the verify account link you can create your new password.
4. Click save.
5. Search login.salesforce.com



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Complete the form to start your free trial. Our team will be in touch to help you make the most of your trial.

First Name*
Your first name

Last Name*
Your last name

Email*
Your email address

Title*
Your job role

Company*
Company Name

Country/Region*
Country/Region

Postal Code*
Your postal code

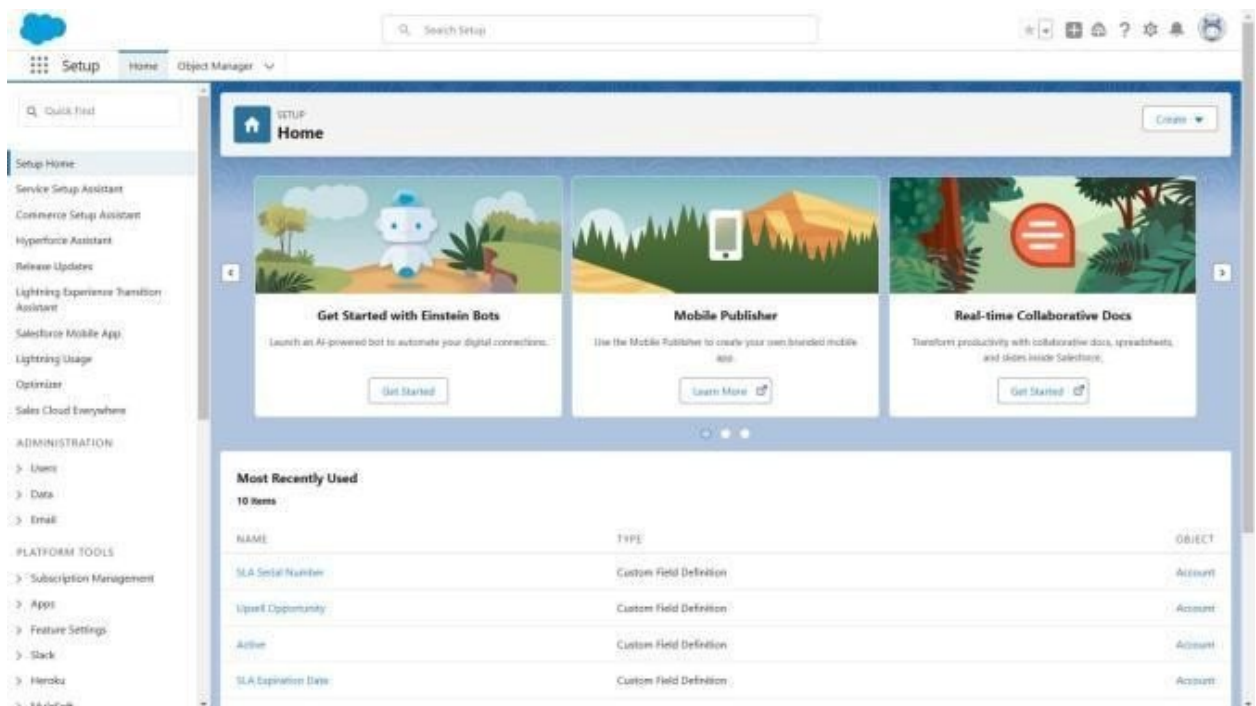
Username*
jane@company.sandbox

Your username must be in the form of an email address (it does not have to be real). It must be unique and cannot be associated with another Salesforce login credential. Read more about username recommendations.

☐ I agree to the Main Services Agreement – Developer Services and Salesforce Program Agreement.

Sign me Up

6. By using username and password you can into the Salesforce org. The homepage will appear below



Setup Home

Get Started with Einstein Bots
Launch an AI-powered bot to automate your digital connections.

Mobile Publisher
Use the Mobile Publisher to create your own branded mobile app.

Real-time Collaborative Docs
Transform productivity with collaborative docs, spreadsheets, and slides inside Salesforce.

Most Recently Used
10 Items

NAME	TYPE	OBJECT
SLA Serial Number	Custom Field Definition	Account
Upsell Opportunity	Custom Field Definition	Account
Action	Custom Field Definition	Account
SLA Expiration Date	Custom Field Definition	Account

Milestone 2-Object:

What is an object?

Salesforce objects are database tables that permit you to store data that is specific to an organization. It consists of fields (columns) and records (rows).

In This Application We Use 4 Custom Objects:

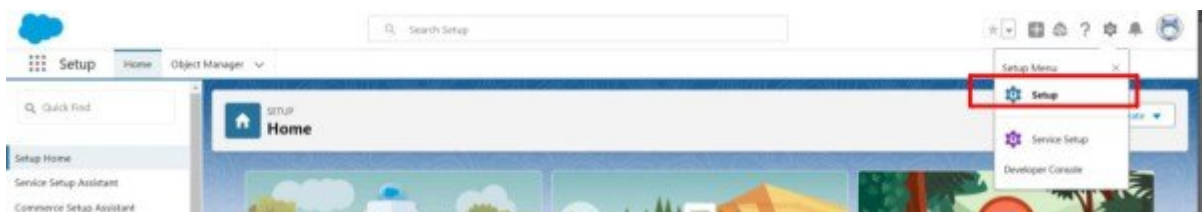
- 1.Farm
- 2.Crop
- 3.Farmer
- 4.Buyer

Activity 1:

Create a custom object for Farm:

To create a custom object, follow these steps:

1. From setup click on object manager.
2. Click create, select custom object.





3. Fill in the label as " Farm ".
4. Fill in the plural label as " Farms".
5. Record name: Farm Name"
6. Select the data type as "Text".

SETUP

New Custom Object

Optional Features

- ☒ Allow Reports
- ☐ Allow Activities
- ☐ Track Field History
- ☐ Allow in Chatter Groups
- ☐ Enable Licensing [i](#)

Object Classification

When these settings are enabled, this object is classified as an Enterprise Application object. When these settings are disabled, this object is classified as a Light Application object. [Learn more.](#)

- ☒ Allow Sharing
- ☒ Allow Bulk API Access
- ☒ Allow Streaming API Access

Deployment Status

☐ In Development

☒ Deployed

Search Status

When this setting is enabled, your users can find records of this object type when they search. [Learn more.](#)

- ☒ Allow Search

Object Creation Options (Available only when custom object is first created)

7. In the Optional Features section, select Allow Reports and Track Field History.
8. In the Deployment Status section, ensure Deployed is selected.
9. In the Search Status section, select Allow Search.
10. In the Object Creation Options section, select Add Notes and Attachments related list to default page layout.
11. Leave everything else as is, and click Save.

Activity 2:

Creation of Crop object:

1. Click on the gear icon and then select Setup.
2. Click on the object manager tab just beside the home tab.
3. After the above steps, have a look on the extreme right you will find a Create Dropdown click on that and select Custom Object.
- a. On the Custom Object Definition page, create the object as follows:

SETUP
New Custom Object

Custom Object Information ⓘ Required Information

The singular and plural labels are used in tabs, page layouts, and reports.

Label Example: Account

Plural Label Example: Accounts

starts with vowel sound ☐

The Object Name is used when referencing the object via the API.

Object Name Example: Account

Description

Context-Sensitive Help Setting ☒ Open the standard Salesforce.com Help & Training window
☐ Open a window using a Visualforce page

Content Name

Enter Record Name Label and Format

The Record Name appears in page layouts, key lists, related lists, lookups, and search results. For example, the Record Name for Account is "Account Name" and for Case it is "Case Number". Note that the Record Name field is always called "Name" when referenced via the API.

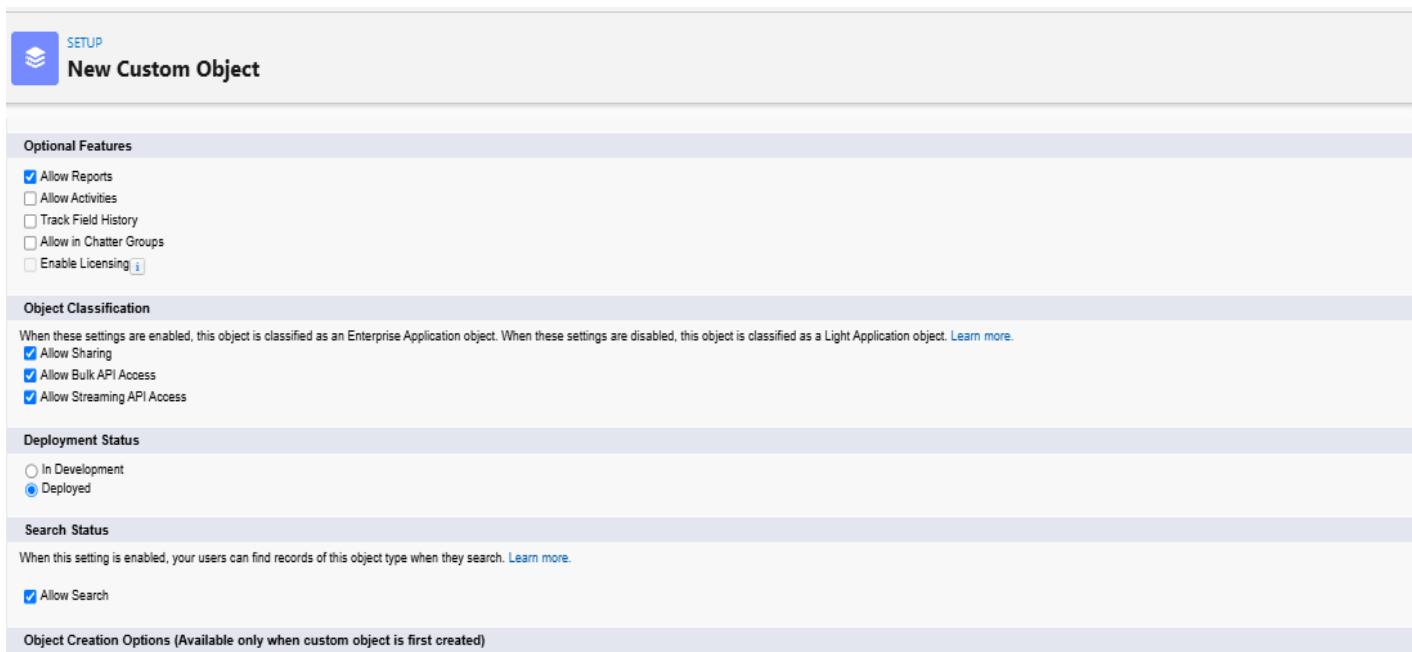
Record Name Example: Account Name

Data Type Warning: If you plan to insert a high volume of records in this object, via the API for example, use the Text data type.

b. Label: Crop

- a. Plural Label: Crops

- b. Record Name: Crop Name
- c. the data type as "Text".



SETUP
New Custom Object

Optional Features

- ☒ Allow Reports
- ☐ Allow Activities
- ☐ Track Field History
- ☐ Allow in Chatter Groups
- ☐ Enable Licensing [i](#)

Object Classification

When these settings are enabled, this object is classified as an Enterprise Application object. When these settings are disabled, this object is classified as a Light Application object. [Learn more.](#)

- ☒ Allow Sharing
- ☒ Allow Bulk API Access
- ☒ Allow Streaming API Access

Deployment Status

☐ In Development
☒ Deployed

Search Status

When this setting is enabled, your users can find records of this object type when they search. [Learn more.](#)

- ☒ Allow Search

Object Creation Options (Available only when custom object is first created)

- d. Check the Allow Reports checkbox
- e. Check the Allow Search checkbox
- f. In the Object Classification section, select Add Notes and Attachments related list to default page layout
- g. Click Save.

Activity 3:

Creation of Farmer object:

1. Click on the gear icon and then select Setup.
2. Click on the object manager tab just beside the home tab.
3. After the above steps, have a look on the extreme right you will find a Create Dropdown click on that and select Custom Object.
 - a. On the Custom Object Definition page, create the object as follows:

SETUP
New Custom Object

Custom Object Information ! Required Information

The singular and plural labels are used in tabs, page layouts, and reports.

Label Example: Account

Plural Label Example: Accounts

Starts with vowel sound ☐

The Object Name is used when referencing the object via the API.

Object Name Example: Account

Description

Context-Sensitive Help Setting ☒ Open the standard Salesforce.com Help & Training window
☐ Open a window using a Visualforce page

Content Name

Enter Record Name Label and Format

The Record Name appears in page layouts, key lists, related lists, lookups, and search results. For example, the Record Name for Account is "Account Name" and for Case it is "Case Number". Note that the Record Name field is always called "Name" when referenced via the API.

Record Name Example: Account Name

Data Type Warning: If you plan to insert a high volume of records in this object, via the API for example, use the Text data type.

b. Label: Farmer

c. Plural Label: Farmers

d.

SETUP
New Custom Object

Optional Features

☒ Allow Reports
☐ Allow Activities
☐ Track Field History
☐ Allow in Chatter Groups
☐ Enable Licensing [i](#)

Object Classification

When these settings are enabled, this object is classified as an Enterprise Application object. When these settings are disabled, this object is classified as a Light Application object. [Learn more.](#)

☒ Allow Sharing
☒ Allow Bulk API Access
☒ Allow Streaming API Access

Deployment Status

☐ In Development
☒ Deployed

Search Status

When this setting is enabled, your users can find records of this object type when they search. [Learn more.](#)

☒ Allow Search

Object Creation Options (Available only when custom object is first created)

Record Name: Farmer Name

e. Select the data type as "Text".

f. Check the Allow Reports checkbox.

g. Check the Allow Search checkbox.

h. In the Object Creation Options section, select Add Notes and Attachments related list to default page layout.

i. Click Save.

Activity 4:

Creation of Buyer object:

1. Click on the gear icon and then select Setup.
2. Click on the object manager tab just beside the home tab.
3. After the above steps, have a look on the extreme right you will find a Create Dropdown click on that and select Custom Object.
- a. On the Custom Object Definition page, create the object as follows:

SETUP
New Custom Object

Custom Object Information Required Information

The singular and plural labels are used in tabs, page layouts, and reports.

Label Example: Account

Plural Label Example: Accounts

starts with vowel sound ☐

The Object Name is used when referencing the object via the API.

Object Name Example: Account

Description

Context-sensitive Help Setting ☒ Open the standard Salesforce.com Help & Training window
☐ Open a window using a Visualforce page

Content Name

Enter Record Name Label and Format

The Record Name appears in page layouts, key lists, related lists, lookups, and search results. For example, the Record Name for Account is "Account Name" and for Case it is "Case Number". Note that the Record Name field is always called "Name" when referenced via the API.

Record Name Example: Account Name

Data Type Warning: If you plan to insert a high volume of records in this object, via the API for example, use the Text data type.

- b. Label: Buyer
- c. Plural Label: Buyers
- d. Record Name: Buyer Name
- e.

SETUP
New Custom Object

Optional Features

☒ Allow Reports
☐ Allow Activities
☐ Track Field History
☐ Allow in Chatter Groups
☐ Enable Licensing

Object Classification

When these settings are enabled, this object is classified as an Enterprise Application object. When these settings are disabled, this object is classified as a Light Application object. [Learn more.](#)

☒ Allow Sharing
☒ Allow Bulk API Access
☒ Allow Streaming API Access

Deployment Status [What is this?](#)

☐ In Development
☒ Deployed

Search Status

When this setting is enabled, your users can find records of this object type when they search. [Learn more.](#)

☒ Allow Search

Object Creation Options (Available only when custom object is first created)

Select the data type as "Text".

- f. Check the Allow Reports checkbox.
- g. Check the Allow Search checkbox.
- h. In the Object Creation Options section, select Add Notes and Attachments related list to default page layout.
- i. Click Save.

Milestone 3 - Tab: Tab

Tabs in Salesforce help users view the information at a glance. It displays the data of objects and other web content in the application.

Activity 1:

Creation of Farm tab:

Now create a custom tab.

1. Click on the Home tab, enter Tabs in Quick Find and select Tabs.

The screenshot shows the Salesforce Setup interface. The top navigation bar includes the Salesforce logo, a search bar labeled "Search Setup", and several utility icons. The left sidebar shows the "Setup" menu with "Home" and "Object Manager" options. The main content area is titled "Tabs" and contains a sub-header "You can create new custom tabs to extend Salesforce functionality or to build new application functionality." Below this, there are four sections: "Custom Object Tabs", "Web Tabs", "Visualforce Tabs", and "Lightning Component Tabs". Each section has a "New" button and a "What Is This?" link. The "Lightning Component Tabs" section is expanded, showing a table with columns for "Action", "Label", "Tab style", and "Description".

Action	Label	Tab style	Description
Edit	Get Started with Agentforce	Agent	
Edit	Get Started with Data Cloud	Map	

2. Under custom object tabs, click New.

 **SETUP**
Tabs

New Custom Object Tab Help for this Page

Step 1. Enter the Details Step 1 of 3

Choose the custom object for this new custom tab. Fill in other details.

New Custom Object Tab Required Information

Select an existing custom object or [create a new custom object now](#).

Object

Tab style

(Optional) Choose a Home Page Custom Link to display on the first time your users click on this tab.

Enter a short description.

--None--

--None--

Buyer

Crop

Farmer

3. For Object, select Farm.
4. For Tab Style, select any icon.
5. Leave all defaults as is. Click Next, Next, and Save.

Activity 2:

Creation of Crop tab:

Now create a custom tab.

1. Click on Home tab, enter Tabs in Quick Find and select Tabs.
2. Under custom object tabs, click New.
3. For Object, select Crop.

The screenshot shows the 'SETUP Tabs' window, specifically 'Step 1. Enter the Details'. The header bar is blue with a gear icon and the text 'SETUP Tabs'. Below the header, a blue bar indicates 'Step 1 of 3'. The main content area is titled 'New Custom Object Tab' and includes a red warning icon with the text 'Required Information'. The instructions state: 'Choose the custom object for this new custom tab. Fill in other details.' The form contains the following fields:

- 'Select an existing custom object or [create a new custom object now.](#)'
- 'Object': A dropdown menu with 'Crop' selected.
- 'Tab Style': A dropdown menu with 'Castle' selected, accompanied by a small icon.
- '(Optional) Choose a Home Page Custom Link to show as a splash page the first time your users click on this tab.' with a 'Splash Page Custom Link' dropdown set to '--None--'.
- 'Enter a short description.' with a 'Description' text area.

At the bottom right, there are 'Next' and 'Cancel' buttons.

4. For Tab Style, select any icon.
5. Leave all defaults as is. Click Next, Next, and Save.

Activity 3:

Creation of Farmer tab:

Now create a custom tab.

1. Click on the Home tab, enter Tabs in Quick Find and select Tabs.
2. Under custom object tabs, click New.

SETUP Tabs

Step 1. Enter the Details Step 1 of 3

Choose the custom object for this new custom tab. Fill in other details.

New Custom Object Tab Required Information

Select an existing custom object or [create a new custom object now](#).

Object: **Farmer**

Tab style: **Dice**

(Optional) Choose a Home Page Custom Link to show as a splash page the first time your users click on this tab.

Splash Page Custom Link: **--None--**

Enter a short description.

Description:

[Next](#) [Cancel](#)

3. For Object, select Farmer.
4. For Tab Style, select any icon.
5. Leave all defaults as is. Click Next, Next, and Save.

Activity 4:

Creation of Buyer tab:

Now create a custom tab.

1. Click on Home tab, enter Tabs in Quick Find and select Tabs.
2. Under custom object tabs, click New.

SETUP Tabs

Step 1. Enter the Details Step 1 of 3

Choose the custom object for this new custom tab. Fill in other details.

New Custom Object Tab Required Information

Select an existing custom object or [create a new custom object now](#).

Object: **Buyer**

Tab style: **Ship**

(Optional) Choose a Home Page Custom Link to show as a splash page the first time your users click on this tab.

Splash Page Custom Link: **--None--**

Enter a short description.

Description:

[Next](#) [Cancel](#)

3. For Object, select Buyer
4. For Tab Style, select any icon.
5. Leave all defaults as is. Click Next, Next, and Save.

Milestone 4 - Lightning App:

What is an App?

Apps in Salesforce area group of tabs that help the application function by working together as a unit. It has a name, a logo, and a particular set of tabs. The simplest app usually has just two tabs.

Activity 1:

Create the Farm Management System app

- From Setup, enter App Manager in the Quick Find and select App Manager.

App Name	Developer Name	Description	Last Modified	App Type	Visibility
2 Agentforce Studio	AgentforceStudio	Agentforce Studio	9/19/2026, 4:27 AM	Lightning	✓
3 All Tabs	AllTabSet		9/19/2026, 4:23 AM	Classic	✓
4 Analytics Studio	Insights	Build CRM Analytics dashboards and apps	9/19/2026, 4:23 AM	Classic	✓
5 App Launcher	AppLauncher	App Launcher tabs	9/19/2026, 4:23 AM	Classic	✓
6 Approvals	Approvals	Manage approvals and approval flows	9/19/2026, 4:23 AM	Lightning	✓
7 Automation	FlowsApp	Automate business processes and repetitive tasks.	9/19/2026, 4:23 AM	Lightning	✓
8 Bolt Solutions	LightningBolt	Discover and manage business solutions designed for your industry.	9/19/2026, 4:23 AM	Lightning	✓
9 Community	Community	Salesforce CRM Communities	9/19/2026, 4:23 AM	Classic	✓
10 Content	Content	Salesforce CRM Content	9/19/2026, 4:23 AM	Classic	✓
11 Data Cloud	Audience360	Build a thorough and complete understanding of your customers.	9/19/2026, 4:23 AM	Lightning	✓
12 Data Manager	DataManager	Use Data Manager to view limits, monitor usage, and manage recipes.	9/19/2026, 4:23 AM	Lightning	✓
13 Developer Edition	Developer_Edition	Welcome to your Developer Edition Org.	9/19/2026, 4:41 AM	Lightning (Managed)	✓
14 DevOps Center	AlmDevOps	Manage your development and release processes easily and collaboratively	9/19/2026, 4:26 AM	Lightning	✓
15 Digital Experiences	SalesforceCMS	Manage content and media for all of your sites.	9/19/2026, 4:23 AM	Lightning	✓

- Click New Lightning App. Farm Management System as the App Name, then click Next
- Under App Options, leave the default selections and click Next.
- Under Utility Items, leave as is and click Next.
- From Available Items, select Farm, Crop, Farmer, Buyer

Reports, and Dashboards and move them to Selected Items.

Navigation Items

Choose the items to include in the app, and arrange the order in which they appear. Users can personalize the navigation to add or move items, but users can't remove or rename the items that you add. Some navigation items are available only for phone or only for desktop. These items are dropped from the navigation bar when the app is viewed in a format that the item doesn't support.

Available Items

🔍 Type to filter list...

📄

AI Prediction Scores

📊

Accounts

🖼️

Action Hub

🎯

Activation Targets

📋

Activations

📅

Activity History

💡

Advisements (Beta)

Create ▼

Selected Items

🏠

Farms

🌾

Crops

👤

Farmers

📄

Buyers

▶

◀

▲

▼

Click Next.

- From Available Profiles, select System Administrator and move it to Selected Profiles.

User Profiles

Choose the user profiles that can access this app.

Available Profiles

🔍 Type to filter list...

Analytics Cloud Integration User

Analytics Cloud Security User

Anypoint Integration

Authenticated Website

Authenticated Website

B2B Reordering Portal Buyer Profile

Contract Manager

Custom: Marketing Profile

▶

◀

Selected Profiles

System Administrator

Click Save & Finish.

- To verify your changes, click the App Launcher, type Farm Management System and select the Job Application app.

Note:

1. App Launcher-Displays available apps.
2. App Name-Displays the current selected app.

3. Navigation menu-Displays the tabs available inside the app.

Milestone 5 - Fields and Relationship: What are fields?

Fields in Salesforce represent what the columns represent in relational databases. It can store data values which are required for a particular object in a record.

Activity 1:

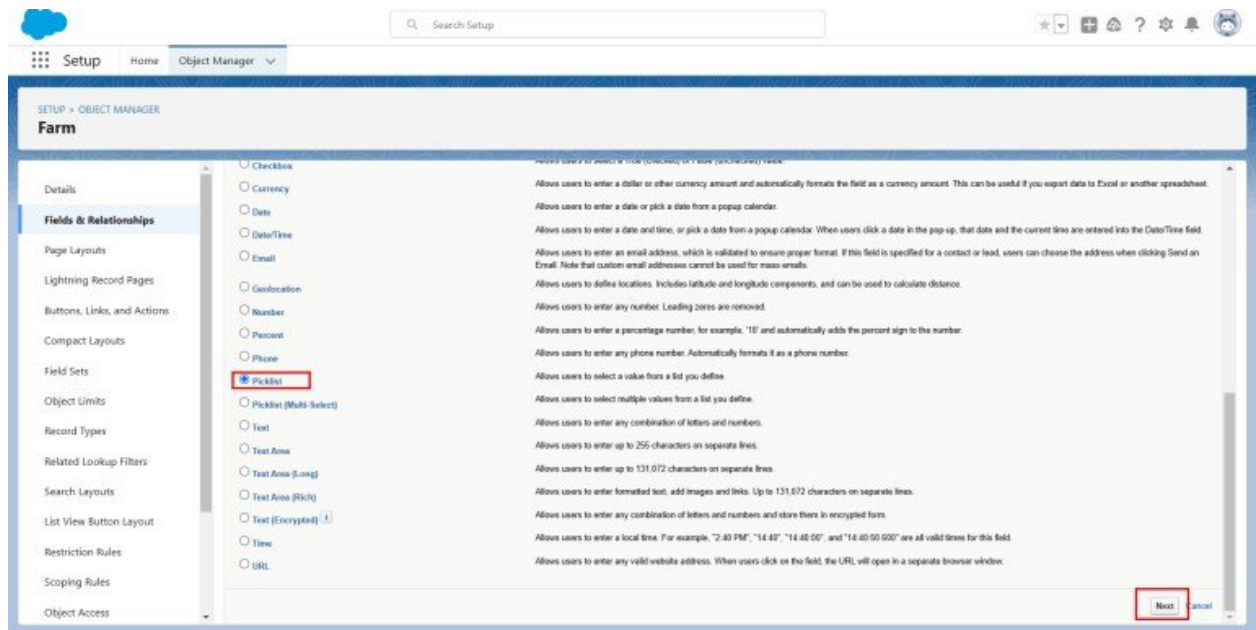
Creation of fields for the Farm objects:

1. click the gear icon and select Setup. This launches Setup in a new tab.
2. Click the Object Manager tab next to Home.
3. Select Farm.
4. Select Fields & Relationships from the left navigation, and click New

The screenshot shows the Salesforce Setup interface for the 'Farm' object. The top navigation bar includes 'Setup', 'Home', and 'Object Manager'. A red arrow points to the 'Object Manager' tab. The left sidebar shows the 'Fields & Relationships' section highlighted. The main content area displays a table of fields for the 'Farm' object. The table has columns for 'FIELD LABEL', 'FIELD NAME', 'DATA TYPE', 'CONTROLLING FIELD', and 'INDEXED'. The 'New' button is highlighted with a red box.

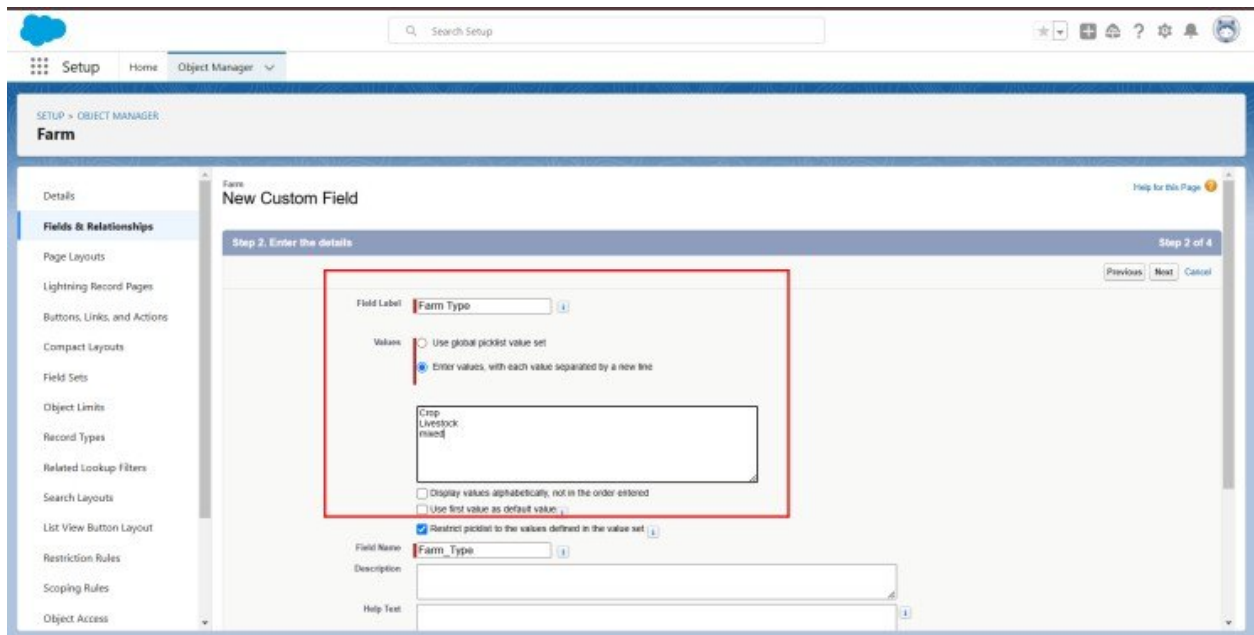
FIELD LABEL	FIELD NAME	DATA TYPE	CONTROLLING FIELD	INDEXED
Created By	CreatedById	Lookup(User)		
Farm Name	Name	Text(80)		✓
Farm Size(Acres)	Farm_Size_Acres__c	Text(30)		
Farm Type	Farm_Type__c	Picklist		
Farmer	Farmer__c	Master-Detail(Farmer)		✓
Irrigation Type	Irrigation_Type__c	Picklist		
Last Modified By	LastModifiedById	Lookup(User)		
Location	Location__c	Text Area(255)		
Soil Quality	Soil_Quality__c	Picklist		

5. Choose Data type as a picklist, Click next



6. Enter Field Label (Farm Type), and Enter Values as (Crop, Livestock, Mixed)

7. Next, Next Click Save.



Now let's create the other fields following Activity1 steps 1 to 4 and we must choose the data types of the fields carefully.

1.Next we Create another Picklist Field,Enter field label (Soil Quality) enter values (Organic,Cover Crops,Composting)

The screenshot shows the configuration page for a picklist field named 'Soil Quality'. At the top, the 'Field Label' is 'Soil Quality'. Below it, a message states: 'Below is a list of picklist values. Click Edit to change the name of the value or make it the default value.' A table lists three values: 'Organic', 'Cover Crops', and 'Composting'. Each row has an 'Action' column with 'Edit' and 'Delete' links, and a 'Modified By' column showing 'Raja G.' and the date '9/24/2026, 2:19 AM'. Below the table, the 'Field Name' is 'Soil_Quality'. There are input fields for 'Description' and 'Help Text'. A 'Required' checkbox is unchecked, with the text 'Always require a value in this field in order to save a record'. A 'Default value' section has a 'Show Formula Editor' link. At the bottom, a small text block explains formula syntax: 'Use formula syntax. Enclose text and picklist value API names in double quotes ("the_text"). Include numbers without quotes (25), show percentages as decimals (0.10), and express date calculations in the standard format: (Today) + 7). To reference a field from a Custom Metadata type record use: \$CustomMetadata.Type__candRecordIDName Field__c'.

Action	Values	API Name	Default	Chart Colors	Modified By
Edit Del	Organic	Organic	☐	Assigned dynamically	Raja G, 9/24/2026, 2:19 AM
Edit Del	Cover Crops	Cover Crops	☐	Assigned dynamically	Raja G, 9/24/2026, 2:19 AM
Edit Del	Composting	Composting	☐	Assigned dynamically	Raja G, 9/24/2026, 2:19 AM

2.Again we Create Picklist field,Field Label (Irrigation type), Enter Values (Sprinkler irrigation,Drip irrigation,localized irrigation)

The screenshot shows the configuration page for a picklist field named 'Irrigation type'. At the top right, there are 'Previous', 'Next', and 'Cancel' buttons. The 'Field Label' is 'Irrigation type'. Under the 'Values' section, the radio button 'Enter values, with each value separated by a new line' is selected. Below this, a text area contains the values: 'Sprinkler irrigation', 'Drip irrigation', and 'localized irrigation'. There are three checkboxes: 'Display values alphabetically, not in the order entered' (unchecked), 'Use first value as default value' (unchecked), and 'Restrict picklist to the values defined in the value set' (checked). Below these, the 'Field Name' is 'Irrigation_type'. There are input fields for 'Description' and 'Help Text'. A 'Required' checkbox is unchecked, with the text 'Always require a value in this field in order to save a record'.

3.Choose data type as a text,enter field label(Location)

Field Label ⓘ

Please enter the maximum length for a text field below.

Length

Field Name ⓘ

Description

Help Text ⓘ

Required ☐ Always require a value in this field in order to save a record

Unique ☐ Do not allow duplicate values

☒ Treat "ABC" and "abc" as duplicate values (case insensitive)

☐ Treat "ABC" and "abc" as different values (case sensitive)

External ID ☐ Set this field as the unique record identifier from an external system

Auto add to custom report type ☒ Add this field to existing custom report types that contain this entity ⓘ

Default Value

Use formula syntax: Enclose text and picklist value API names in double quotes : ("the_text"). Include numbers without quotes : (25), show percentages as decimals: (0.10), and express date calculations in the standard format: (Today() + 7). To reference a field from a Custom Metadata type record use: \$CustomMetadata.Type__mdt.RecordAPIName.Field__c

4.Choose data type as text,enter field label (size)

Field Label ⓘ

Please enter the maximum length for a text field below.

Length

Field Name ⓘ

Description

Help Text ⓘ

Required ☐ Always require a value in this field in order to save a record

Unique ☐ Do not allow duplicate values

☒ Treat "ABC" and "abc" as duplicate values (case insensitive)

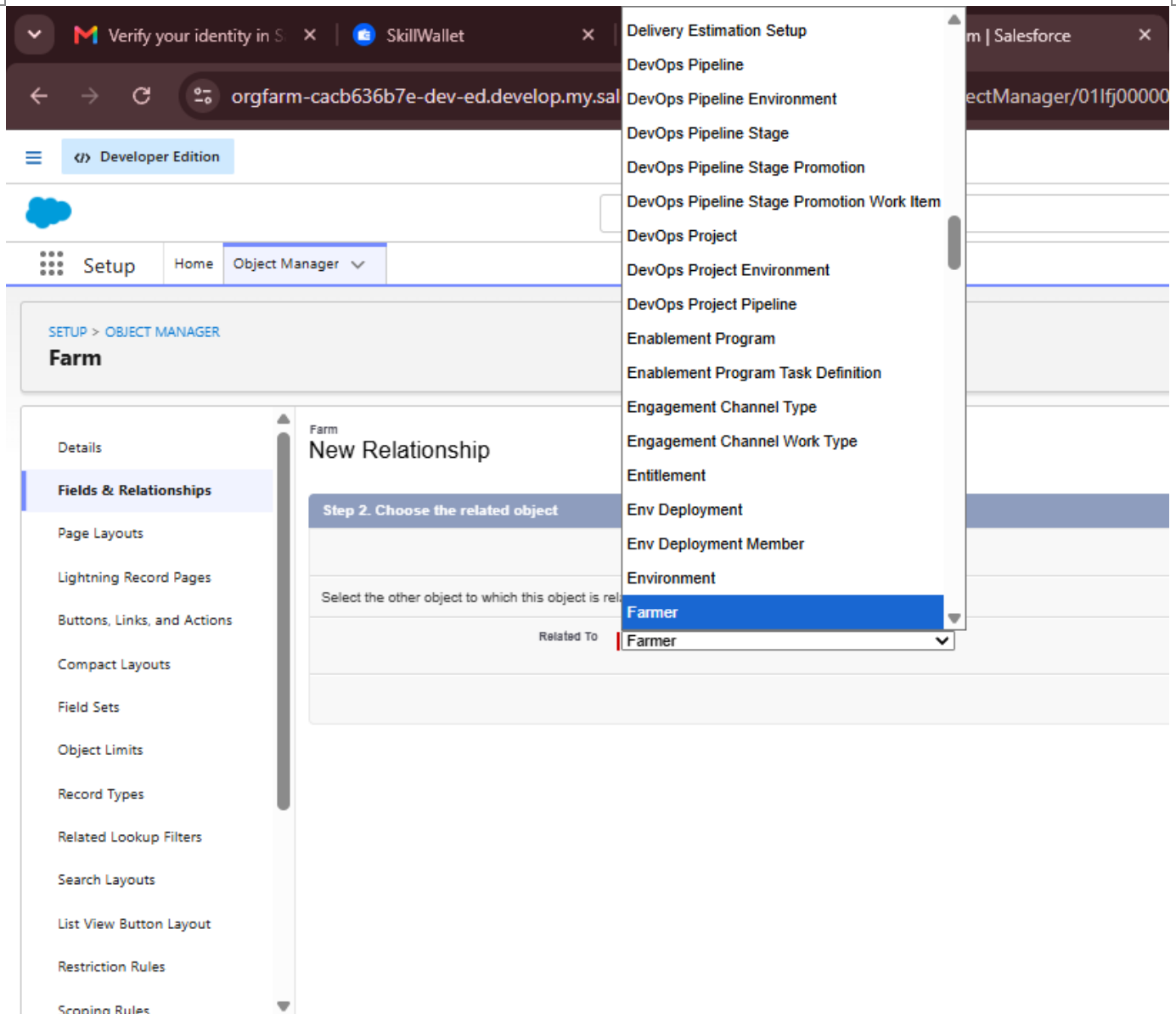
☐ Treat "ABC" and "abc" as different values (case sensitive)

External ID ☐ Set this field as the unique record identifier from an external system

Auto add to custom report type ☒ Add this field to existing custom report types that contain this entity ⓘ

5.Create a Master Detail relationship with Farmer Object.

☐ Auto Number	A system-generated sequence number that uses a display format you define. The number is automatically incremented for each new record.
☐ Formula	A read-only field that derives its value from a formula expression you define. The formula field is updated when any of the source fields change.
☐ Roll-Up Summary 	A read-only field that displays the sum, minimum, or maximum value of a field in a related list or the record count of all records listed in a related list.
☐ Lookup Relationship	Creates a relationship that links this object to another object. The relationship field allows users to click on a lookup icon to select a value from a popup list. The other object is the source of the values in the list.
☒ Master-Detail Relationship	Creates a special type of parent-child relationship between this object (the child, or "detail") and another object (the parent, or "master") where: • The relationship field is required on all detail records. • The ownership and sharing of a detail record are determined by the master record. • When a user deletes the master record, all detail records are deleted. • You can create rollup summary fields on the master record to summarize the detail records. The relationship field allows users to click on a lookup icon to select a value from a popup list. The master object is the source of the values in the list.
☐ External Lookup Relationship	Creates a relationship that links this object to an external object whose data is stored outside the Salesforce org.
☐ Checkbox	Allows users to select a True (checked) or False (unchecked) value.
☐ Currency	Allows users to enter a dollar or other currency amount and automatically formats the field as a currency amount. This can be useful if you export data to Excel or another spreadsheet.
☐ Date	Allows users to enter a date or pick a date from a popup calendar.
☐ Date/Time	Allows users to enter a date and time, or pick a date from a popup calendar. When users click a date in the pop-up, that date and the current time are entered into the Date/Time field.
☐ Email	Allows users to enter an email address, which is validated to ensure proper format. If this field is specified for a contact or lead, users can choose the address when clicking Send an Email. Note that custom email addresses cannot be used for mass emails.
☐ Geolocation	Allows users to define locations. Includes latitude and longitude components, and can be used to calculate distance.
☐ Number	Allows users to enter any number. Leading zeros are removed.
☐ Percent	Allows users to enter a percentage number, for example, '10' and automatically adds the percent sign to the number.



a)click next,next,save.

Activity 2: Creation of fields for the Crop objects:

1.Choose Data type As a text,Enter Field Label name (Crop Name),

Custom Field Definition Edit
Change Field Type
Save
Cancel

Field Information
Data Type Text
Required Information

Field Label Crop Name
Field Name Crop_Name
Description
Help Text
Data Owner User
Field Usage --None--
Data Sensitivity Level --None--
Compliance Categorization
Available
PII
HIPAA
GDPR
PCI
Chosen

General Options

Next save.

2.Select Data type as date ,Enter Field label (Planting Date),then save .

Crop
New Custom Field

Step 2. Enter the details

Field Label Planting Date
Field Name Planting_Date
Description
Help Text
Required
Always require a value in this field in order to save a record
Auto add to custom report type
Add this field to existing custom report types that contain this entity
Default Value
Hide Formula Editor
Select Field Type
-- Select field type --
Insert Field
-- Insert Merge Field --
Use Formula Assistant
Insert Operator

3.Again we select data type as a date,Enter field labe (Harvest date),save

New Custom Field

Step 2. Enter the details

Field Label ⓘ

Field Name ⓘ

Description

Help Text ⓘ

Required ☐ Always require a value in this field in order to save a record

Auto add to custom report type ☒ Add this field to existing custom report types that contain this entity ⓘ

Default Value [Hide Formula Editor](#)

Select Field Type Insert Field [Use Formula Assistant](#) Insert Operator

4.Next We Create Lookup relationship on crop object realted to Farm Object.

Specify the type of information that the custom field will contain.

Data Type

☐ None Selected Select one of the data types below.

☐ Auto Number A system-generated sequence number that uses a display format you define. The number is automatically incremented for each new record.

☐ Formula A read-only field that derives its value from a formula expression you define. The formula field is updated when any of the source fields change.

☐ Roll-Up Summary ⓘ A read-only field that displays the sum, minimum, or maximum value of a field in a related list or the record count of all records listed in a related list.

☒ Lookup Relationship Creates a relationship that links this object to another object. The relationship field allows users to click on a lookup icon to select a value from a popup list. The other object is the source of the values in the list.

☐ Master-Detail Relationship Creates a special type of parent-child relationship between this object (the child, or "detail") and another object (the parent, or "master") where:

- The relationship field is required on all detail records.
- The ownership and sharing of a detail record are determined by the master record.
- When a user deletes the master record, all detail records are deleted.
- You can create rollup summary fields on the master record to summarize the detail records.

The relationship field allows users to click on a lookup icon to select a value from a popup list. The master object is the source of the values in the list.

☐ External Lookup Relationship Creates a relationship that links this object to an external object whose data is stored outside the Salesforce org.

☐ Checkbox Allows users to select a True (checked) or False (unchecked) value.

☐ Currency Allows users to enter a dollar or other currency amount and automatically formats the field as a currency amount. This can be useful if you export data to Excel or another spreadsheet.

☐ Date Allows users to enter a date or pick a date from a popup calendar.

5.We select adata type as currency,Enter field Label (Actual Yield), Next Save it.

Step 2. Enter the details

Step 2 of 2

PreviousNextCancel

Field LabelActual Yield

Please enter the length of the number and the number of decimal places. For example, a number with a length of 8 and 2 decimal places can accept values up to "12345678.90".

Length18

Number of digits to the left of the decimal point

Decimal Places0

Number of digits to the right of the decimal point

Field NameActual_Yield

Description

Help Text

Required

☐ Always require a value in this field in order to save a record

Auto add to custom report type

☒ Add this field to existing custom report types that contain this entity

Default Value

Show Formula Editor

Use formula syntax: Enclose text and picklist value API names in double quotes ("the text"), include numbers without quotes. (2017) shows percentages as decimals (0.10), and express date calculations in the standard format: Today() + 7). To reference a field from a Custom Metadata type record use: \$CustomMetadata.Type__mdt.RecordAPIName.Field__c

6. Select Data type as a Number, enter Field Label (Total Acres Harvested), Enter Maximum Length (20) and save.

New Custom Field

Help for this F

Step 2. Enter the details

Step

PreviousNext

Field LabelTotal Acres Harvested

Please enter the length of the number and the number of decimal places. For example, a number with a length of 8 and 2 decimal places can accept values up to "12345678.90".

Length17

Number of digits to the left of the decimal point

Decimal Places0

Number of digits to the right of the decimal point

Field NameTotal_Acres_Harvested

Description

Help Text

Required

☐ Always require a value in this field in order to save a record

Unique

☐ Do not allow duplicate values

External ID

☐ Set this field as the unique record identifier from an external system

AI Prediction

☐ Use this field to store AI prediction scores

Auto add to custom report type

☒ Add this field to existing custom report types that contain this entity

Default Value

Show Formula Editor

Use formula syntax: Enclose text and picklist value API names in double quotes ("the text"), include numbers without quotes. (2017) shows percentages as decimals (0.10), and express date calculations in the standard format: Today() + 7). To reference a field from a Custom Metadata type record use: \$CustomMetadata.Type__mdt.RecordAPIName.Field__c

7. Now We create Formula Field For That Select data Type As a Formula, Click Next.

8. Enter Field Label (Expected Yield per Acre), Select formula return type as Currency, click Next.

Step 2. Choose output type

Field Label: Expected Yield per Acre

Field Name: Expected_Yield_per_Acre

Auto add to custom report type: ☒ Add this field to existing custom report types that contain this entity

Formula Return Type

☐ None Selected

Select one of the data types below.

☐ Checkbox

☒ Currency

☐ Date

☐ DateTime

☐ Number

☐ Percent

☐ Text

☐ Time

Options

Decimal Places: 2 Example: 999.00

Previous Next

9. Enter a Formula Actual Yield Divide By Total Acre Harvested, then click next and save.

Step 3. Enter formula

Enter your formula and click Check Syntax to check for errors. Click the Advanced Formula subtab to use additional fields, operators, and functions.

Example: `[Gross Margin = Amount - Cost_c]` [More Examples...](#)

Simple Formula Advanced Formula

Insert Field

Use Formula Assistant

Insert Operator

Expected Yield per Acre (Currency) =

`Actual_Yield__c / Total_Acres_Harvested__c`

Check Syntax

Functions

-- All Function Categories --

ABS

ACOS

ADDMONTHS

AND

ASCII

ASIN

Insert Selected Function

Quick Tips

- Getting Started
- Operators & Functions

Activity 3: Creation of fields for the Farmer objects:

1. Select Data type Phone, Enter field label (Contact Number), next Save.

NEW CUSTOM FIELD

Step 2. Enter the details

Field Label

Field Name

Description

Help Text

Required ☐ Always require a value in this field in order to save a record

Auto add to custom report type ☒ Add this field to existing custom report types that contain this entity

Default Value

Use formula syntax. Enclose text and picklist value API names in double quotes ("The text"). Include numbers without quotes. (2+3) refers to the sum of 2 and 3, and express date calculations in the standard format: (Today() + 7). To reference a field from a Custom Metadata type record use: \$CustomMetadata.Type__c.myRecordAPIName.field__c

2. Select Data type Picklist, Enter Field Label (State), Enter values (Telangana, Andhra Pradesh, Tamilnadu), save.

Previous **Next**

Field Label

Values ☐ Use global picklist value set

☒ Enter values, with each value separated by a new line

☐ Display values alphabetically, not in the order entered

☐ Use first value as default value

☒ Restrict picklist to the values defined in the value set

Field Name

Description

Help Text

Required ☐ Always require a value in this field in order to save a record

Auto add to custom report type ☒ Add this field to existing custom report types that contain this entity

Default Value

Use formula syntax. Enclose text and picklist value API names in double quotes ("The text"). Include numbers without quotes. (2+3) refers to the sum of 2 and 3, and express date calculations in the standard format: (Today() + 7). To reference a field from a Custom Metadata type record use: \$CustomMetadata.Type__c.myRecordAPIName.field__c

3. Choose Data type Picklist, Enter Field Label (District), Enter Values of district, next save

Step 2. Enter the details

Field Label:

Values: ☐ Use global picklist value set
☒ Enter values, with each value separated by a new line

☐ Display values alphabetically, not in the order entered
☐ Use first value as default value
☒ Restrict picklist to the values defined in the value set

Field Name:

Description:

Help Text:

4. Again select Data type as Picklist, Enter Field Label (Village), Enter Values of villages, next save.

Step 2. Enter the details

Field Label:

Values: ☐ Use global picklist value set
☒ Enter values, with each value separated by a new line

☐ Display values alphabetically, not in the order entered
☐ Use first value as default value
☒ Restrict picklist to the values defined in the value set

Field Name:

Description:

Help Text:

Required: ☐ Always require a value in this field in order to save a record
Auto add to custom report type: ☒ Add this field to existing custom report types that contain this entity

Default Value:

Creation of field dependency :

1. Now create Field Dependency For District field, Open Object Manager, Choose Farmer object, select Fields & Relationships, On Right side We have Field Dependency option, select that option.

2. Click new, Select Controlling Field as State, Dependent Field As District, Click on Continue.

1. Select the appropriate Value in each column by double-clicking them.
2. For Ex. Telangana - Yadadri Bhongir.

5. Click Include Values. And it is also the same for Andhra Pradesh, Tamilnadu with its city.

6. Click Preview, then test the dependency by selecting a different State and viewing the associate Values available for Particular state.

The screenshot shows the Salesforce 'Setup' interface with the 'Object Manager' tab selected. The left sidebar lists various setup options, with 'Fields & Relationships' highlighted. The main content area is titled 'New Field Dependency' and provides instructions on creating a dependent relationship. It includes two steps: Step 1 (Select a controlling field and a dependent field) and Step 2 (Edit filter rules). A red box highlights the 'Controlling Field' and 'Dependent Field' dropdown menus, both of which currently show 'None--'. The 'Continue' and 'Cancel' buttons are visible next to each dropdown.

Setup > OBJECT MANAGER
Farmer

Details

Fields & Relationships

Page Layouts

Lightning Record Pages

Buttons, Links, and Actions

Compact Layouts

Field Sets

Object Limits

Record Types

Related Lookup Filters

Search Layouts

List View Button Layout

Restriction Rules

Scoping Rules

Object Access

New Field Dependency

Create a dependent relationship that causes the values in a picklist or multi-select picklist to be dynamically filtered based on the value selected by the user in another field.

- The field that drives filtering is called the "controlling field." Standard and custom checkboxes and picklists with at least one and less than 300 values can be controlling fields.
- The field that has its values filtered is called the "dependent field." Custom picklists and multi-select picklists can be dependent fields.

Step 1. Select a controlling field and a dependent field. Click Continue when finished.

Step 2. On the following page, edit the filter rules that control the values that appear in the dependent field for each value in the controlling field.

Controlling Field	None--	Continue	Cancel
Dependent Field	None--	Continue	Cancel



Edit Field Dependency

[Help for this Page](#)[Save](#) [Cancel](#) [Preview](#)Controlling Field: State
Dependent Field: District

▼ instructions

- Double click on a cell to toggle its visibility for the Controlling Field value shown in the column heading.
- To change multiple cells at once, select multiple cells and then click the Include Values or Exclude Values button to change the visibility of all selected cells at once.
- Use SHIFT + click to select a range of adjacent cells.
- Double click on a cell to include its value in the dependent picklist.
- Use the Preview button to test the results.

Legend

Excluded Value

Included Value

Click button to include or exclude selected values from the dependent picklist:

[Include Values](#)[Exclude Values](#)

Showing Columns: 1 - 3 (of 3) < Previous Next > View All Go to	
State:	Telangana
District:	Thiruvarur
	Thiruvannamalai
	Thiruvannamalai
	Thiruvannamalai
Showing Columns: 1 - 3 (of 3) < Previous Next > View All	

Click button to include or exclude selected values from the dependent picklist:

[Include Values](#)[Exclude Values](#)[Save](#) [Cancel](#) [Preview](#)

7. Click Close to close the preview window, Click Save.

Activity 4:

Creation of fields for the Buyer objects:

1. Select Data type as a Phone, Enter a Field Label(Contact), Save.

Farm
New Custom Field

Step 2. Enter the details

Field Label

Field Name

Description

Help Text

Required ☐ Always require a value in this field in order to save a record

Auto add to custom report type ☒ Add this field to existing custom report types that contain this entity

Default Value [Show Formula Editor](#)

Use formula syntax. Enclose text and picklist value API names in double quotes ("the text"). Include numbers without quotes. Use percent signs as decimals (0.10), and express date calculations in the standard format: Today() + 7. To reference a field from a Custom Metadata type record use \$CustomMetadataType__c.recordId/FieldName__c.

2. Select Data type as an Email, Enter Field Label (Email), Next save.

Edit Farm Custom Field
Email

Custom Field Definition Edit

Change Field Type Save Cancel

Field Information

Field Label: Email

Field Name: Email

Description:

Help Text:

Data Owner: User

Field Usage: --None--

Data Sensitivity Level: --None--

Compliance Categorization

Available: PII, HIPAA, GDPR, PCI

Chosen:

General Options

Required: ☐ Always require a value in this field in order to save a record

Unique: ☐ Do not allow duplicate values

External ID: ☐ Set this field as the unique record identifier from an external system

Default Value: Show Formula Editor

Use formula syntax. Enclose text and picklist value API names in double quotes ("The API"). Include numbers without quotes. (25% above percentages as decimals (0.10), and express date calculations in the standard format: (today() + 7). To reference a field from a Custom Metadata type record use: \$CustomMetadata.Type__mdt.RecordAPIName.Field__c

Change Field Type Save Cancel

3. Select Data type Text area, Enter Field Label (Address), Save it.

Step 2. Enter the details

Field Label: Address

Please enter the maximum length for a text field below.

Length: 50

Field Name: Address

Description:

Help Text:

Required: ☐ Always require a value in this field in order to save a record

Unique: ☐ Do not allow duplicate values

External ID: ☐ Set this field as the unique record identifier from an external system

Auto add to custom report type: ☒ Add this field to existing custom report types that contain this entity

Default Value: Show Formula Editor

Use formula syntax. Enclose text and picklist value API names in double quotes ("The API"). Include numbers without quotes. (25% above percentages as decimals (0.10), and express date calculations in the standard format: (today() + 7). To reference a field from a Custom Metadata type record use: \$CustomMetadata.Type__mdt.RecordAPIName.Field__c

4. Select Data type Picklist, Field Label (Buyer Type), Enter values (Wholesale, Retail), Click Save.

Step 2: Enter the details Step

Field Label 1

Values ☐ Use global picklist value set
☒ Enter values, with each value separated by a new line

☐ Display values alphabetically, not in the order entered
☐ Use first value as default value 1
☒ Restrict picklist to the values defined in the value set 1

Field Name 1

Description

Help Text

Required ☐ Always require a value in this field in order to save a record
Auto add to custom report type ☒ Add this field to existing custom report types that contain this entity 1

Default Value 1

Previous Next

Milestone 6- Profile:

What is a profile?

A profile is a group/collection of settings and permissions that define what a user can do in

salesforce. A profile controls "Object permissions, Field permissions, User permissions, Tab

settings, App settings, Apex class access, Visualforce page access, Page layouts, Record Types, Login hours & Login IP ranges.

A profile can be assigned to many users, but user can be assigned single profile at a time.

Activity 1:

Create a custom profile:

1. From setup , enter profiles in Quick Find box
2. Select profiles (Standard user).
3. Click clone.
4. For Profile, enter Sales Department.





SETUP

Profiles

If your organization uses Record Types, use the Edit links in the Record Type Settings section below to make one or more record types available to users with this profile.

[Login IP Ranges](#) | [Enabled Apex Class Access](#) | [Enabled Visualforce Page Access](#) | [Enabled External Data Source Access](#) | [Enabled Named Credential Access](#) | [Enabled External Credential Provider Access](#) | [Enabled Custom Metadata Type Access](#) | [Enabled Custom Setting Definitions Access](#) | [Enabled Flow Access](#) | [Enabled Service Presence Status Access](#) | [Enabled Custom Permissions](#)

Profile Detail

[Edit](#) [Clone](#) [View Users](#)

Name	Standard User		
User License	Salesforce	Custom Profile	☐
Created By	Salesforce, Inc., 9/19/2026, 4:23 AM	Modified By	Rag G. 9/24/2026, 8:25 AM

Page Layouts

Standard Object Layouts			
Global	Global Layout View Assignment	Location Group Assignment	Location Group Assignment Layout View Assignment
Email Application	Not Assigned View Assignment	Macro	Macro Layout View Assignment
Home Page Layout	Home Page Default View Assignment	Object Milestone	Object Milestone Layout View Assignment
Account	Account Layout View Assignment	Operating Hours	Operating Hours Layout View Assignment
Alternative Payment Method	Alternative Payment Method Layout View Assignment	Opportunity	Opportunity Layout View Assignment
Appointment Invitation	Appointment Invitation Layout View Assignment	Opportunity Product	Opportunity Product Layout View Assignment
Asset	Asset Layout View Assignment	Order	Order Layout View Assignment
Asset Action	Asset Action Layout View Assignment	Order Product	Order Product Layout View Assignment
Asset Action Source	Asset Action Source Layout View Assignment	Payment	Payment Layout View Assignment
Asset Relationship	Asset Relationship Layout View Assignment	Payment Authorization	Payment Authorization Layout View Assignment
Asset State Period	Asset State Period Layout View Assignment	Payment Authorization Adjustment	Payment Authorization Adjustment Layout View Assignment
Assigned Resource	Assigned Resource Layout View Assignment	Payment Gateway	Payment Gateway Layout View Assignment

anager ▾



Clone Profile

[Help for this I](#)

Enter the name of the new profile.

You must select an existing profile to clone from.

Required information

Existing Profile	Standard User
User License	Salesforce
Profile Name	Sales Department

Click save.

Activity 2:

Create a custom profile-2:

Create a profile with the profile name as “ Farm Manager” .

5. From setup , enter profiles in Quick Find box
6. Select profiles (Standard user).



The screenshot shows the Salesforce Setup interface for Profiles. At the top, there's a header with a user icon, the word "SETUP" in blue, and "Profiles" in bold. Below this, the section is titled "Clone Profile" with a subtitle "Enter the name of the new profile." A modal dialog box is open with the heading "You must select an existing profile to clone from." Inside the dialog, there's a table with two columns: "Existing Profile" and "User License". The first row shows "Standard User" and "Salesforce". Below the table, there's a "Profile Name" label followed by a text input field containing "Farm Manager". At the bottom right of the dialog are "Save" and "Cancel" buttons.

Existing Profile	User License
Standard User	Salesforce

Profile Name:

Save Cancel

7. Click clone.

Milestone 7- Role

In Salesforce, roles are used to determine which users have access to certain data and functions within the system. They are also used to define the reporting hierarchy within an organization. Users with higher roles have greater access to data and more control over the system.

Activity-1 Creation of role

1. From the Quick find box search for the role and click on the roles option
2. select the set up roles option

SETUP Roles

Role Edit
New Role

Role Edit

Label Farm Manager

Role Name Farm Manager

This role reports to CEO

Role Name as displayed on reports

Save Save & New Cancel

Milestone 8-User :

What is a user?

A user is anyone who logs into Salesforce. Users are employees at your company, such as sales reps, managers, and IT specialists, who need access to the company's records. Every user in

Salesforce has a user account. The user account identifies the user, and the user account settings determine what features and records the user can access.

NOTE- As Salesforce license can only be used by 2 Users at a time in Dev Org, so If you don't find salesforce license then deactivate a user who has salesforce license Or change the license type from Salesforce to any other.

Activity 1:

To Create a user:

1. From Setup, enter Users in the Quick Find box, then select Users.
2. Click New User.

SETUP Users

All Users

On this page you can create, view, and manage users.

To get more licenses, use the Your Account app. [Let's Go](#)

View: **All Users** [Edit](#) [Create New View](#)

Action	Full Name	Alias	Username	Role	Active	Profile
Edit	Chatter Expert	Chatter	chatt000f00000hvgbes2.xmswmbasou@chatter.salesforce.com		✓	Chatter Free User
Edit	EPIC OnFarm	OEPIG	epic-9e1e18b6166@onfarm.salesforce.com		✓	System Administrator
Edit	G. Raja	gan	ganesanb61.7f89047f1384@gagentforce.com		✓	System Administrator
Edit	User Integration	info	integrations000f00000hvgbes2.com		✓	Analytics Cloud Integration User
Edit	User Security	sec	insightsecurity000f00000hvgbes2.com		✓	Analytics Cloud Security User

[New User](#) [Reset Password\(s\)](#) [Add Multiple Users](#)

- Enter First name as Farm and last name as Manager.
- Enter the user's name and email address and a unique username in the form of an email address. By default, the username is the same as the email address.
- Select the role Farm Manager.
- Select user License as Standard Platform User.
- Select profile(Farm Manager).

SETUP Users

User Edit

[Cancel](#) [Save](#) [Save & New](#)

General Information

First Name	Farm	Role	None Specified
Last Name	Manager	User License	Salesforce
Alias	fmana	Profile	Standard User
Email	ganesanb61@gmail.com	Active	☒
Username	ganesanb61.7f89047f1384	Marketing User	☐
Nickname	User179031706682833222	Offline User	☐
Title		Knowledge User	☐
Company		Flow User	☐
Department		Service Cloud User	☐
Division		Site.com Contributor User	☐
		Site.com Publisher User	☐
		WDC User	☐
		Delta.com User Type	None
		Delta.com Monthly Addition Limit	Default Limit (300)
		Accessibility Mode (Classic Only)	☐
		High-Contrast Palette on Charts	☐
		Load Lightning Pages While Scrolling	☒
		Debug Mode	☐
		Quick Access Menu	☒
		Salesforce CRM Content User	☒
		Receive Salesforce CRM Content Email Alerts	☒
		Receive Salesforce CRM Content Alerts as Daily Digest	☒
		Allow Forecasting	☐
		No MRU Updates	☐
		Call Center	

- Click save.

Activity 2:

Create a user with a username as “Sales Department”, and assign him the sales Manager profile. Follow the steps from above Activity

SETUP Users

New User

Salesforce now sends emails only from verified domains. To avoid delivery failures, verify your email domains or enable your substitute email-sending domain. [Learn More](#)

User Edit [Cancel] [Save] [Save & New]

General information

First Name	Farm
Last Name	Manager
Alias	fmana
Email	ganesanb81@gmail.com
Username	Sales Department
Nickname	User179031791040378440
Title	
Company	
Department	
Division	

Role Farm Manager

User License Salesforce Platform

Profile Standard Platform User

Active ☒

Marketing User ☐

Offline User ☐

Knowledge User ☐

Flow User ☐

Service Cloud User ☐

Site.com Contributor User ☐

Site.com Publisher User ☐

WDC User ☐

Data.com User Type --None--

Data.com Monthly Addition Limit Default Limit (300)

Accessibility Mode (Classic Only) ☐

High-Contrast Palette on Charts ☐

Load Lightning Pages While Scrolling ☒

Debug Mode ☐

Quick Access Menu ☒

Salesforce CRM Content User ☒

Receive Salesforce CRM Content Email Alerts ☒

Milestone 9-Sharing Rules:

What are Sharing Rules?

Sharing rules help users to share records based on conditions. It is basically created for objects whose organization-wide defaults (OWD) are set to public read-only or private because sharing rules can only extend the access and not restrict it.

Types of sharing rules,

- Owner-based Sharing Rules
- Criteria-based Sharing Rules

Activity 1:

Create a sharing rule

- 1) Goto Sharing Settings, which can be found under the Quick Find section.
- 2) Scroll down and find the candidate object where a sharing rule needs to be added,

and then click on New to create a new sharing rule.

- 3) Add the label of the sharing rule you want to make.
- 4) Select your rule type based on the criteria.
- 5) Select the field can join immediately check field from the Crop object.
- 6) Select the Crop as equal and value is True.
- 7) And in selecting the users to share with the section select roles and in that select Farm Manager.
- 8) And in the section of select the level of access for the users give the access Read/Write.
- 9) And save the rule.

SS

try

SETUP

Sharing Settings

Note: "Roles and subordinates" includes all users in a role, and the roles below that role.

You can use sharing rules only to grant wider access to data, not to restrict access.

Step 1: Rule Name

Label

Crop

Rule Name

Crop

Description

Step 2: Select your rule type

Rule Type

☐ Based on record owner ☒ Based on criteria

Step 3: Select which records to be shared

Criteria

Field	Operator	Value	
Crop Name	equals	True	AND
--None--	--None--		AND
--None--	--None--		AND
--None--	--None--		AND
--None--	--None--		AND

[Add Filter Logic...](#)

Additional Options

☒ Include records owned by users who can't have an assigned role

Step 4: Select the users to share with

Share with

Roles

Farm Manager

Step 5: Select the level of access for the users

Access Level

Read/Write

Save

Cancel

Activity 2:

Create a Sharing Rule to Share the records of Buyer to Sales department with the Access of Read/Write.

Follow the steps from above Activity.

The screenshot shows the 'Sharing Settings' setup page in Salesforce. It includes a 'SETUP' icon and a 'Sharing Settings' title. A note states: 'Note: "Roles and subordinates" includes all users in a role, and the roles below that role. You can use sharing rules only to grant wider access to data, not to restrict access.'

Step 1: Rule Name

Label: Buyer
Rule Name: Buyer
Description:

Step 2: Select your rule type

Rule Type: ☐ Based on record owner ☒ Based on criteria

Step 3: Select which records to be shared

Criteria:

Field	Operator	Value	
Buyer Name	equals	True	AND
--None--	--None--		AND
--None--	--None--		AND
--None--	--None--		AND
--None--	--None--		

Add Filter Logic...

Additional Options: ☒ Include records owned by users who can't have an assigned role

Step 4: Select the users to share with

Share with: Roles Farm Manager

Step 5: Select the level of access for the users

Access Level: Read/Write

Buttons: Save Cancel

Milestone-10: User Adoption

Activity 1: Create a Record(Farmer) • Click on App Launcher on left side of screen.

- Search Farm Management & click on it.
- Click on Farmer Tab.
- Click new and fill details & Save.

New Farm

* = Required Information

Information

* Farm Name 

Green Valley Farm

Farm Type

--None--

Soil Quality

--None--

Irrigation type

--None--

Location

Size

Cancel Save & New Save

Milestone 11-Reports:

What are Reports?

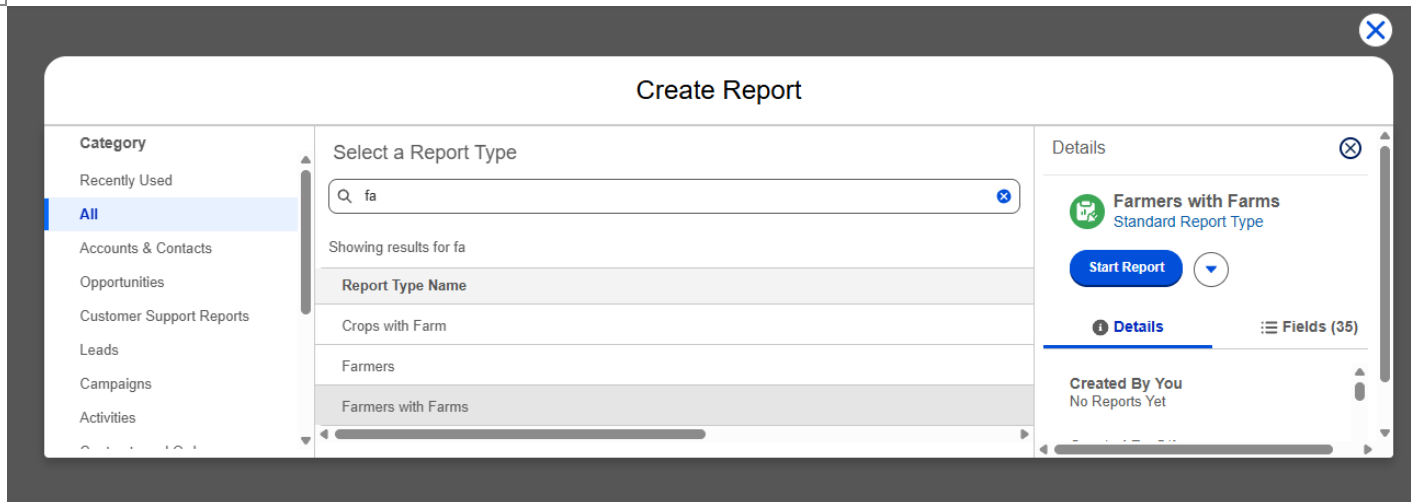
A report is a list of records that meet the criteria you define. It's displayed in rows and columns, and can be filtered, grouped, or displayed in a graphical chart. Every report is stored in a folder. Folders can be public, hidden, or shared, and can be set to read-only or read/write.

Activity 1:

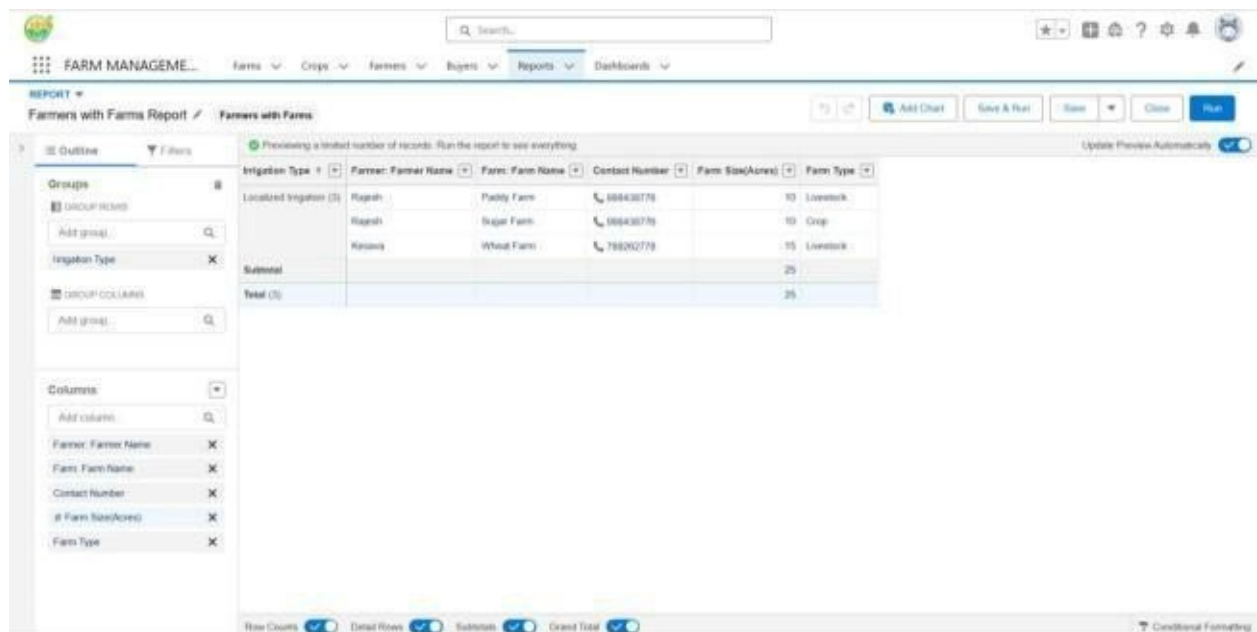
Create a report:

Create a report that displays rating of the account and which has type and account name.

- 1) Click on app launcher search for reports.
- 2) Click on the new report and select the category has Farmer with Farms.

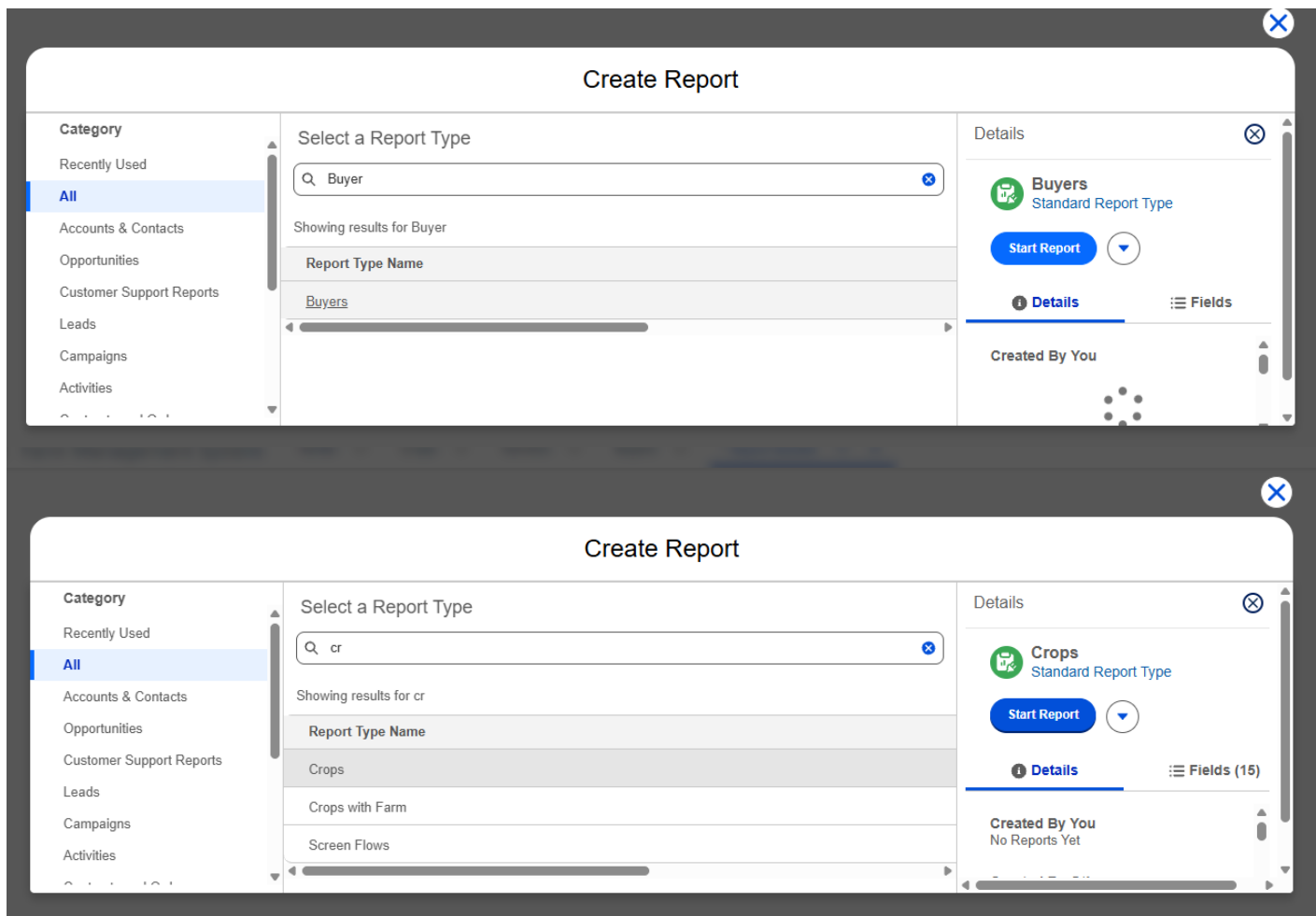


- 3) In the details section select the option start report.
- 4) In Outline Click on Column Select Which Field you want to display in the Report.
- 5) Click Group,Select Which Fields You Want to Group.
- 6) Save the report by giving label name(Farmer with Farms) and save the folder as a public folder and save the report.



Activity 2:

Create a Report using the Objects Crops, Buyers. Follow the steps from above Activity.



Create a Dashboard:

1. Click the Dashboard tab.
2. Click New Dashboard.
3. Name the dashboard Farmers with farms and click Create.
4. Click +Component.
5. Select the Farmers with Farms and click Select.
6. Select the Gauge chart and click Add.
7. Click Save and then Done.

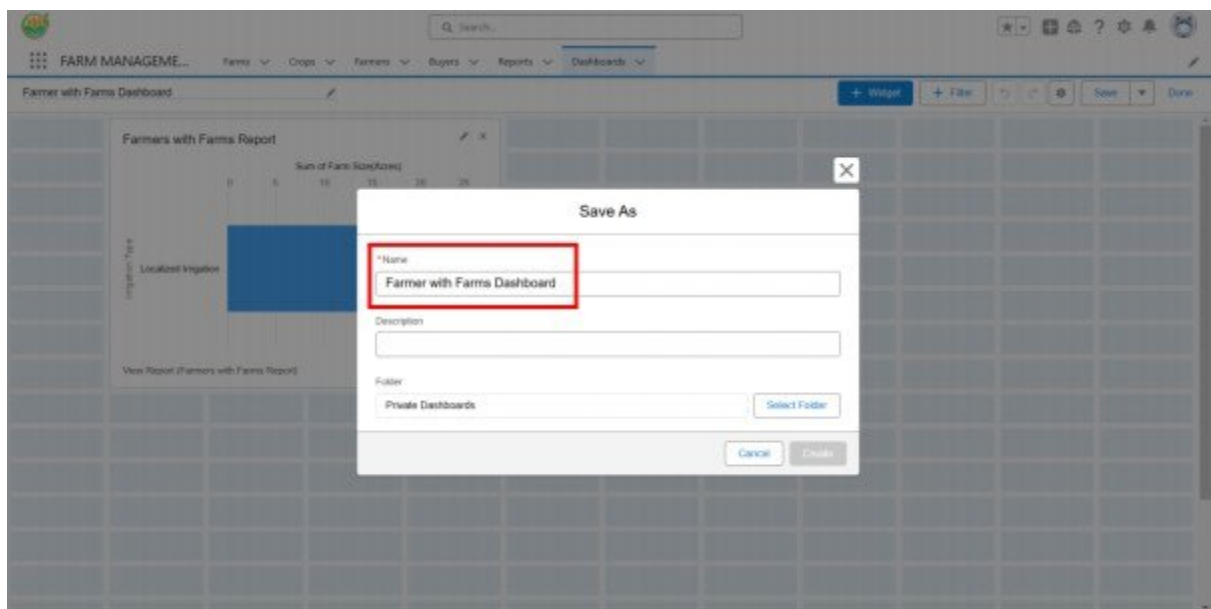
Milestone 12-Dashboard:

Dashboards provide more insights than reports as they combine the data from many reports and show a summarized result. Looking at many reports at a time gives the

flexibility of combining the results from them quickly. Also summaries in dashboards help us decide on action plans quicker. The dashboards can contain charts, graphs and Tabular data.

Activity 1: Create a Dashboard:

8. Click the Dashboard tab.
9. Click New Dashboard.
10. Name the dashboard Farmers with farms and click Create.
11. Click +Component.
12. Select the Farmers with Farms and click Select.
13. Select the Gauge chart and click Add.
14. Click Save and then Done.





Search...

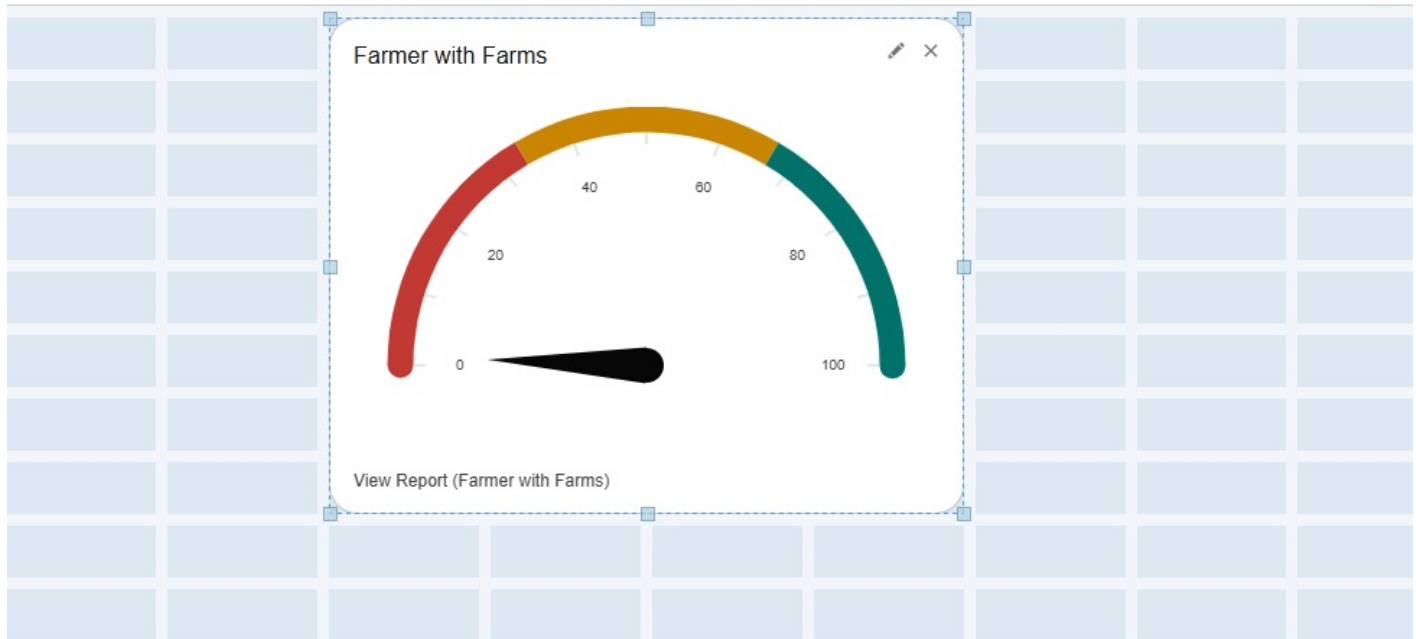


Developer Edition

Welcome

Farmers with farms Dashboard

+ Widget



Milestone-13: View Report and Dashboard:

Activity 1: Report

- Click on App Launcher on left side of screen.
- Search Farm Management System & click on it.
- Click on Reports Tab.
- Click on Farmers with Farms & see records



Search...



Farm Management System

Farms

Crops

Farmer

Buyers

* Farmer with Farms



Report: Farmer with Farms
Farmer with Farms

Enable Field Editing



Add Chart



Edit

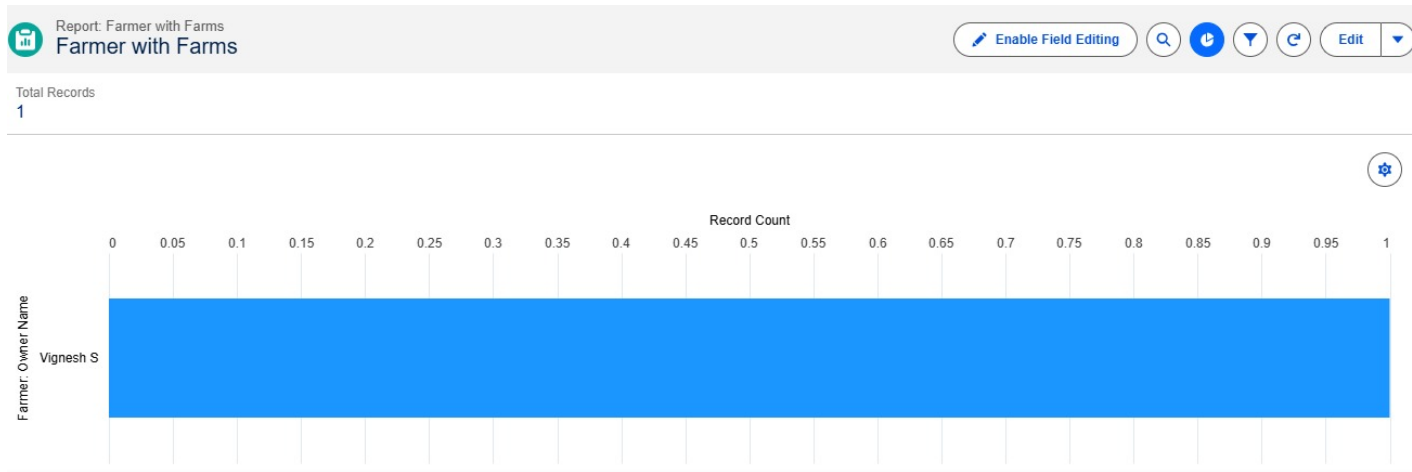


Total Records
1

Farmer: Owner Name	Farmer: Farmer Name	vignesh	Total
☐ Vignesh S	Record Count	1	1
Total	Record Count	1	1

Activity 2: Dashboard

- Click on App Launcher on left side of screen.
- Search Farm Management system & click on it.
- Click on Dashboard Tab.
- Click Farmers With farms & see records.



7. FUNCTIONAL & PERFORMANCE TESTING

During the testing phase, screenshots were captured from the Salesforce interface to verify system functionality. The following were tested:

- Creation of Farm, Crop, Farmer, and Buyer records
- Validation of input fields
- Relationship and dependency checks
- Profile and role access testing
- Sharing rule verification
- Report and dashboard generation

Performance testing ensured smooth system response and stability during multiple user operations.

8. ADVANTAGES & DISADVANTAGES

Advantages

- Centralized data system
- Reduced paperwork
- Improved accuracy
- Better monitoring
- Real-time analytics

Disadvantages

- Requires internet connection
- Salesforce license dependency
- Limited offline access

9. Conclusion:

The Farm Management System successfully digitizes agricultural data management using Salesforce. It improves efficiency, reduces errors, and enables farmers and managers to make informed decisions through automation and analytics.

10. FUTURE SCOPE

- Mobile application integration
- AI-based crop prediction
- Weather data integration
- IoT sensor connectivity
- Government scheme integration
- Market price forecasting

11. APPENDIX

- Source Code: Salesforce Configuration and Metadata
- Dataset: Farm, Crop, Farmer, and Buyer Records

- Tools Used: Salesforce Developer Org

Thank You

